

Anneal–Neutral Estimation of Hard-Max Dynamic Discrete Choice Models*

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Abstract

Dynamic discrete choice models with deterministic decision rules generate non-smooth objective functions because optimal policies switch at indifference boundaries. Standard approaches embed fixed logit smoothing into the behavioural model, but this introduces pseudo-true bias when the data-generating process is hard-max. This paper proposes an Anneal-Neutral Estimator (ANE) that treats log-sum-exp smoothing as a purely computational device: an annealed likelihood is maximised at each sample size while the temperature parameter $\tau_n \rightarrow 0$ ensures the limiting estimator targets the hard-max model. Under a transversality condition, indifference sets form lower-dimensional manifolds with zero probability under the invariant distribution—the measure-zero indifference (MZI) property. Consequently, the population log-likelihood is twice differentiable at the true parameter despite pointwise kinks in the Bellman operator. We show that ANE is $\sqrt{n}$ -consistent and asymptotically normal, with the same limiting variance as the infeasible hard-max maximum likelihood estimator. The framework accommodates neural-network sieve approximation of value functions, extending the estimator to high-dimensional state spaces. Monte Carlo experiments in a dynamic labour supply model confirm that ANE recovers accurate policy thresholds, whereas permanently smoothed estimators exhibit substantial bias.

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1 Introduction

1.1 Motivation

Many economic decisions are governed by hard thresholds. A worker facing a kinked budget constraint works zero hours below a reservation wage and positive hours above it; a firm replaces a machine once cumulative maintenance costs cross a critical level; a household adjusts its durable stock only when the gap between desired and actual holdings exceeds a transaction cost. In each case the optimal decision rule is deterministic: the agent selects the action that maximises the expected present value of payoffs, with no continuously distributed taste shock entering at the moment of choice.

Such hard-max behaviour is a central feature of leading dynamic discrete choice (DDC) applications. Labour supply and welfare participation respond sharply to benefit eligibility thresholds and non-convexities in tax–benefit schedules (Hotz and Miller, 1993; Keane and Wolpin, 1997). Equipment replacement policies follow (S, s) -type rules (Rust, 1987; Cooper and Haltiwanger, 2006). Export entry and exit decisions exhibit persistent inaction regions generated by sunk costs (Das et al., 2007; Ryan, 2012). In all these environments the economic primitives deliver kinked value functions and policies with genuine switching boundaries.

Structural estimation of DDC models has traditionally sidestepped the hard-max structure by embedding smoothing directly into the behavioural model. The most common approach, introduced by Rust (1987), adds i.i.d. extreme-value shocks to each action’s payoff, yielding a logit conditional choice probability controlled by a scale parameter. When this scale parameter is fixed at a positive value, the smoothing is a maintained behavioural assumption: agents are modelled as making stochastic rather than deterministic choices.

Two difficulties arise when the data-generating process is in fact hard-max. First, the estimator targets a pseudo-true parameter that minimises the Kullback–Leibler divergence between the smoothed model and the true hard-max process; the resulting bias is concentrated near the policy threshold and can be economically significant. Second, the smoothed model misspecifies the shape of choice probabilities near the threshold, distorting counterfactual policy predictions even when the threshold location is recovered approximately.¹

This paper asks whether one can retain the computational benefits of log-sum-exp smoothing while estimating the hard-max structural model consistently. We propose an Anneal-Neutral Estimator (ANE) that uses temperature-indexed smoothing as a purely computational device—not a behavioural assumption—by letting the temperature vanish as the sample size grows. Because the smoothing disappears, ANE targets the true hard-max parameter rather than a pseudo-true approximation. Because the smoothing is positive at every finite sample, the optimisation problem remains smooth and amenable to gradient-based solvers.

1.2 Overview of the Approach

ANE combines three ingredients: vanishing temperature smoothing, a geometric property of the hard-max model, and neural-network sieve approximation of value functions.

Vanishing temperature smoothing. At each positive temperature the Bellman operator and its associated conditional choice probabilities are smooth functions of both the structural parameter and the value function. We solve a sequence of smoothed problems at decreasing temperatures,

¹Manski (1985) notes a related point for the maximum score estimator: the distinction between smooth and non-smooth objectives has first-order consequences for both the rate of convergence and the form of the limiting distribution.

each initialised at the previous solution. The final temperature depends on the sample size and vanishes at a rate that ensures the estimator converges to the hard-max parameter without inflating the asymptotic variance. The temperature plays the same role as the vanishing bandwidth in nonparametric simulated maximum likelihood (Kristensen and Shin, 2012) or the temperature in Laplace-type estimators (Chernozhukov and Hong, 2003): it regularises finite-sample optimisation without altering the large-sample target.

Measure-zero indifference (MZI). The population log-likelihood of the hard-max model is smooth in the structural parameter even though the Bellman operator and the hard-max policy are point-wise non-differentiable. The reason is geometric: under a transversality condition on value-function differences, the set of states at which two or more actions are simultaneously optimal—the indifference set—forms a lower-dimensional manifold that has measure zero under any absolutely continuous state distribution. Integration over states therefore averages out the kinks, and the population criterion inherits the smoothness of its integrand away from the indifference set. The formal statement and proof appear in Section 3.

Neural-network sieve approximation. For models with high-dimensional state spaces, grid-based value iteration becomes infeasible. We approximate the value function by a feedforward neural network whose depth and width grow with the sample size, exploiting approximation rates from Yarotsky (2017) and Farrell et al. (2021). The sieve approximation error enters the asymptotic theory through a rate condition that balances approximation against the smoothing parameter. Section 5 develops the details.

1.3 Relationship to Existing Literature

1.3.1 Smoothing in DDC Models

Rust (1987) introduced the extreme-value specification yielding the log-sum-exp integrated value function. Aguirregabiria and Mira (2002, 2007) developed the Nested Pseudo-Likelihood (NPL) algorithm, which uses CCP estimates to avoid full value-function computation but maintains a fixed smoothing parameter embedded in the behavioural model. In the static discrete choice literature, McFadden (1989) developed logit-smoothed simulators for multinomial probit, and Iskhakov et al. (2017) extended smoothing methods to dynamic models with taste shocks and continuous states.

The Neural-Network Efficient Estimator (NNES) of Nguyen (2026) illustrates how neural value-function approximations can be incorporated in smooth-logit DDC models while preserving root- n efficiency through a zero-Jacobian structure.

Most closely related to this paper, Kristensen, Mogensen, Moon, and Schjerning (2021) propose solving DDC models using smoothed Bellman operators combined with polynomial sieve approximations. They replace the max operator with a log-sum-exp, develop an asymptotic theory for the smoothed sieve solutions, and show convergence as the smoothing vanishes. Their framework shares several features with ours: both use log-sum-exp smoothing with a vanishing parameter, both employ sieve approximations, and both establish convergence to the true value function. This paper differs in three ways. First, their focus is on solving the model—approximating the value function for a given parameter—whereas ours is on estimating the structural parameter via maximum likelihood. Second, we provide the MZI characterisation that explains why the hard-max population criterion is smooth, which is not needed for their value-function problem but is essential for the estimation theory. Third, we use neural-network rather than polynomial sieves, which offer superior scaling in high dimensions. Table 1 summarises the comparison.

Kristensen and Shin (2012) develop nonparametric simulated maximum likelihood estimation where a vanishing kernel bandwidth plays the same theoretical role as our vanishing temperature.

Table 1: Comparison with Kristensen, Mogensen, Moon, and Schjerning (2021)

	KMMS (2021)	This paper
Goal	Solve for V_θ given θ	Estimate θ_0
Smoothing	Log-sum-exp, vanishing	Log-sum-exp, vanishing
Sieve	Polynomials	Neural networks
Main result	$\ V_{\theta,\tau}^{(K)} - V_\theta\ = O(\cdot)$	$\sqrt{n}(\hat{\theta}_n - \theta_0) \Rightarrow N$
MZI / indifference geometry	Not needed	Central to the argument
State-space scaling	$O(K^{d_s})$	$O(d_s \cdot K)$

Their bandwidth conditions are analogues of our rate conditions, specialised to continuous-choice likelihood simulation rather than the DDC Bellman structure. The DDC setting introduces the additional complication of value-function approximation error.

1.3.2 Alternative Computational Approaches

Su and Judd (2012) propose Mathematical Programming with Equilibrium Constraints (MPEC) for DDC estimation. MPEC imposes the Bellman equation as a constraint rather than solving it as an inner loop, and is highly efficient for low-dimensional state spaces. However, it requires a discrete grid whose size scales exponentially in the state dimension. ANE and MPEC are complementary: MPEC is preferable for small state spaces where sparse nonlinear programming solvers are fast, while ANE with neural sieves extends to high-dimensional problems.

Conditional choice probability methods (Hotz and Miller, 1993; Arcidiacono and Miller, 2011) avoid solving the Bellman equation by inverting nonparametrically estimated CCPs. These methods require full-support choice probabilities—a condition that fails in the hard-max model where choice probabilities take values in $\{0, 1\}$ almost everywhere. ANE avoids this difficulty because annealed CCPs have full support for any positive temperature.

Bajari et al. (2007) propose a two-step simulation-based estimator for dynamic games that also requires smooth choice probabilities. Our approach preserves the structural Bellman equations and targets the hard-max model, using annealed smoothing and neural sieves to restore differentiability.

1.3.3 Non-Smooth M-Estimation

The general theory of M-estimation with non-smooth objectives is well developed. Pakes and Pollard (1989) provide conditions under which $\sqrt{n}$ asymptotics hold for discontinuous objective functions. Kim and Pollard (1990) show that when the population criterion has a non-quadratic maximum (e.g., a kink), the convergence rate slows to $n^{1/3}$ with a non-Gaussian limit. In our setting, the population criterion is smooth at the true parameter (by the MZI property), so the cube-root case does not arise. The $\sqrt{n}$ rate follows from standard MLE theory (Newey and McFadden, 1994) once the smoothness of the population criterion is established; the MZI property is the DDC-specific result that verifies this smoothness.

Chernozhukov and Hong (2003) study Laplace-type estimators with a temperature parameter for computational purposes and show that the temperature does not affect the limiting distribution provided it vanishes at an appropriate rate. The parallel to our annealing neutrality result is direct. We specialise this idea to the DDC Bellman structure, where the value-function fixed point requires explicit treatment of the approximation error.

1.3.4 Neural-Network Sieves

Farrell et al. (2021) establish approximation rates and inference results for deep neural networks in semiparametric models. Yarotsky (2017) derives sharp approximation rates for ReLU networks. Related work shows that kinks along lower-dimensional manifolds do not preclude fast approximation (Petersen and Voigtlaender, 2018; Imaizumi and Fukumizu, 2019). The Neural-Network Efficient Estimator (NNES) of Nguyen (2026) illustrates how neural value-function approximations can be incorporated in smooth-logit DDC models while preserving root- n efficiency through a zero-Jacobian structure. We use neural networks to approximate the value function at each annealing stage; the approximation rate depends on the smoothness of the value function away from the kink set, and the kink contributes only through a logarithmic correction (Section 5).

1.4 Contributions

1. *Measure-zero indifference (MZI) property.* We show that the indifference set is a $(d_s - 1)$ -dimensional manifold with measure zero under the invariant distribution (Lemma 3.5). The population log-likelihood is therefore differentiable at the true parameter despite point-wise kinks in the Bellman operator. While the principle that integration smooths lower-dimensional singularities is classical, the contribution is to verify the required transversality condition within the DDC Bellman fixed-point structure, handling the implicit dependence of the value function on the structural parameter through the parameterised-contraction framework of Bonnans and Shapiro (2000).
2. *Annealing neutrality.* The annealed estimator converges to the same parameter as the infeasible hard-max MLE (Proposition 4.7), and the vanishing temperature does not affect the asymptotic distribution (Theorem 6.3). The conceptual idea that vanishing computational smoothing preserves asymptotics is known from Kristensen and Shin (2012) and Chernozhukov and Hong (2003); the contribution is to develop the result within the DDC Bellman structure, where the approximation error and the discount factor introduce additional rate conditions.
3. *$\sqrt{n}$ -consistency and asymptotic normality.* We establish $\sqrt{n}(\hat{\theta}_n - \theta_0) \Rightarrow N(0, H_0^{-1} \Omega H_0^{-1})$ (Theorem 6.3), where H_0 and Ω are the Hessian and score variance of the hard-max population criterion. The proofs include verification of the Lyapunov condition for the triangular-array score under margin condition $\kappa > 2$, and a uniform law of large numbers for the Hessian under the double limit in which both the parameter estimate and the temperature converge.
4. *Feasible rate conditions.* We derive explicit conditions on the annealing rate, the sieve approximation rate, and the margin exponent under which all required rate conditions hold simultaneously (Proposition 5.4), and show that grid-based methods avoid the sieve constraint entirely.
5. *Scalable implementation.* The annealing framework combined with neural-network sieve approximation of the value function extends the estimator to state spaces with four or more dimensions, where grid-based methods become infeasible.
6. *Monte Carlo evidence.* Experiments in a dynamic labour supply model confirm that ANE recovers accurate policy thresholds, while permanently smoothed estimators exhibit substantial bias (Section 8).

1.5 Outline

Section 2 sets up the DDC model, defines the indifference set, and states the identification assumption. Section 3 develops the measure-zero indifference property and its implications for the smoothness of the population criterion. Section 4 introduces the annealed Bellman operator, proves convergence as the temperature vanishes, and establishes annealing neutrality. Section 5 presents the neural-network sieve approximation and derives feasible rate conditions. Section 6 contains the main asymptotic results. Section 7 describes the computational algorithm. Section 8 reports Monte Carlo evidence. Appendix A contains proofs of the main results, Appendix B the asymptotic theory proofs, and Appendix C auxiliary results.

2 Model and Setup

This section sets up the dynamic discrete choice model, defines the indifference set and its geometry, discusses the hard-max likelihood and why direct maximisation is infeasible, and states the identification and regularity assumptions.

2.1 The Dynamic Discrete Choice Problem

An infinitely-lived agent observes a state $s_t \in \mathcal{S}$ at each period $t = 0, 1, 2, \dots$ and chooses an action $a_t \in \mathcal{A} = \{0, 1, \dots, J\}$ to maximise the expected discounted sum of per-period payoffs. The state space $\mathcal{S} \subseteq \mathbb{R}^{d_s}$ is assumed to be a compact, convex set with nonempty interior.² The action space $\mathcal{A}$ is finite.

The per-period utility is $u(s, a; \theta)$, where $\theta \in \Theta \subset \mathbb{R}^{d_\theta}$ is a vector of structural parameters and Θ is compact. The state evolves according to a controlled Markov transition kernel $F(\cdot | s, a)$, which assigns a probability measure on $(\mathcal{S}, B(\mathcal{S}))$ for each $(s, a) \in \mathcal{S} \times \mathcal{A}$. We assume throughout that the Markov process admits an invariant distribution μ_θ for each θ and that μ_{θ_0} is unique at the true parameter.

Hard-max decision rule. We consider environments in which the agent’s optimal policy is a deterministic (hard-max) rule:

$$a^*(s; \theta) = \operatorname{argmax}_{a \in \mathcal{A}} Q_\theta(s, a; V_\theta), \quad (1)$$

where Q_θ is the action-value (choice-specific value) function and V_θ is the value function. Crucially, *no additive choice-specific i.i.d. taste shock* enters the decision at the moment of choice. This distinguishes the present setting from the standard Rust (1987) framework, where extreme-value shocks are part of the structural model, and from the NPL framework of Aguirregabiria and Mira (2002), which requires smooth CCPs.

The action-value function satisfies the Bellman equation

$$Q_\theta(s, a; V) = u(s, a; \theta) + \beta \int_{\mathcal{S}} V(s') F(ds' | s, a), \quad (2)$$

where $\beta \in [0, 1)$ is the discount factor, and the value function is the fixed point of the Bellman operator T_θ :

$$V_\theta(s) = (T_\theta V_\theta)(s) := \max_{a \in \mathcal{A}} Q_\theta(s, a; V_\theta). \quad (3)$$

²Compactness is used in the contraction mapping argument for the Bellman equation and in the uniform convergence arguments of Section 6. In the Monte Carlo (Section 8), we truncate the state space and discuss the implications of truncation for boundary behavior.

Under the compactness and boundedness conditions stated below (Assumption 5), T_θ is a contraction on the space $B(\mathcal{S})$ of bounded measurable functions equipped with the sup-norm, and V_θ exists and is unique by the Banach fixed-point theorem (e.g. Stokey et al., 1989).

Remark 2.1 (Notation). We write V_θ for the value function $V(\cdot; \theta)$ to emphasise its dependence on θ ; the notation $V_{\theta, \tau}$ will be used from Section 4 onward for the annealed value function at temperature $\tau > 0$. We use $F(ds' | s, a)$ throughout to denote the transition kernel in measure-theoretic notation, which accommodates both continuous densities and mixed discrete-continuous transitions. When F admits a density $f(s' | s, a)$ with respect to Lebesgue measure λ on $\mathcal{S}$, we write $F(ds' | s, a) = f(s' | s, a) ds'$.

2.2 Indifference Geometry

The hard-max policy (1) is uniquely defined at all states s where one action is strictly better than all others, but may be set-valued at states where two or more actions are tied. We define the *optimal correspondence*

$$\Gamma_\theta(s) := \operatorname{argmax}_{a \in \mathcal{A}} Q_\theta(s, a; V_\theta) \quad (4)$$

and the *indifference set* (also called the kink set, since the value function V_θ is non-differentiable precisely at these points):

Definition 2.2 (Indifference set). The indifference set at parameter θ is

$$\mathcal{K}(\theta) := \{s \in \mathcal{S} : |\Gamma_\theta(s)| \geq 2\} = \{s \in \mathcal{S} : \exists a \neq a' \text{ with } a, a' \in \Gamma_\theta(s)\}.$$

The indifference set consists of states at which two or more actions are *simultaneously optimal*. This is distinct from the set $\{s : \Delta_{a, a'}(s; \theta) = 0\}$ for a given pair (a, a') , where $\Delta_{a, a'}(s; \theta) := Q_\theta(s, a; V_\theta) - Q_\theta(s, a'; V_\theta)$ is the value-function difference. The latter set may include states where neither a nor a' is optimal (because a third action a'' dominates both); at such states the value function has no kink, and including them in $\mathcal{K}(\theta)$ would be incorrect.³ For models with two actions ($\mathcal{A} = \{0, 1\}$), the indifference set simplifies to $\mathcal{K}(\theta) = \{s : \Delta_{0,1}(s; \theta) = 0\}$, and the distinction above is immaterial.

To characterise the geometry of $\mathcal{K}(\theta)$, it is convenient to work with pairwise value differences. For any $(a, a') \in \mathcal{A}^2$ and parameter θ , define

$$\Delta_{a, a'}(s; \theta) \equiv Q_\theta(s, a; V_\theta) - Q_\theta(s, a'; V_\theta), \quad s \in \mathcal{S}. \quad (5)$$

States at which $\Delta_{a, a'}(s; \theta) = 0$ are states at which the agent is indifferent between actions a and a' . Because the value function solves the fixed-point equation (3), the map $s \mapsto \Delta_{a, a'}(s; \theta)$ embeds the implicit dependence of V_θ on θ .

Smoothness of value differences. A key observation is that although V_θ itself has kinks at $\mathcal{K}(\theta)$, the action-value *differences* $\Delta_{a, a'}$ are smooth. To see why, write

$$Q_\theta(s, a; V_\theta) - Q_\theta(s, a'; V_\theta) = u(s, a; \theta) - u(s, a'; \theta) + \beta \int V_\theta(s') [F(ds' | s, a) - F(ds' | s, a')].$$

³**Example.** Suppose $\mathcal{A} = \{0, 1, 2\}$ with $Q_\theta(s, 0; V) = Q_\theta(s, 1; V) = 0$ and $Q_\theta(s, 2; V) = 1$ at some state s . Then $\Delta_{0,1}(s; \theta) = 0$ but $\Gamma_\theta(s) = \{2\}$, so $s \notin \mathcal{K}(\theta)$. The value function $V_\theta(s) = 1$ is smooth at s because the max is uniquely attained.

The function $s \mapsto \int V_\theta(s')F(ds'|s, a)$ inherits the smoothness of the transition kernel in s , provided F has a smooth density: the kinks of V_θ in s' are integrated out by the absolute continuity of $F(\cdot|s, a)$ with respect to Lebesgue measure. Thus the differentiability of $\Delta_{a,a'}$ in s depends on the smoothness of u and F , but not on the differentiability of V_θ itself.

Transversality. To ensure that $\mathcal{K}(\theta)$ is a lower-dimensional manifold (rather than a thick set), we impose a transversality condition on the value-function differences.

Assumption 1 (Transversality). *For every pair $a \neq a'$ such that $\{s : \Delta_{a,a'}(s; \theta_0) = 0, a, a' \in \Gamma_{\theta_0}(s)\} \neq \emptyset$:*

- (i) *The function $s \mapsto \Delta_{a,a'}(s; \theta_0)$ is continuously differentiable on a neighbourhood of $\mathcal{K}_{a,a'}(\theta_0) := \{s : a, a' \in \Gamma_{\theta_0}(s)\}$ in $\mathcal{S} \setminus (\mathcal{K}(\theta_0) \setminus \mathcal{K}_{a,a'}(\theta_0))$.*
- (ii) *The gradient is nonvanishing on the indifference surface: $\nabla_s \Delta_{a,a'}(s; \theta_0) \neq 0$ for all $s \in \mathcal{K}_{a,a'}(\theta_0)$.*
- (iii) *The indifference set is contained in the interior: $\mathcal{K}(\theta_0) \subset \text{int}(\mathcal{S})$.*

Condition (i) requires $\Delta_{a,a'}$ to be C^1 in a neighbourhood of the relevant indifference surface; by the argument above, this holds whenever u and F are sufficiently smooth. Condition (ii) is the standard transversality condition from differential topology: together with (i), it allows us to apply the implicit function theorem and the regular value theorem (Guillemin and Pollack, 1974) to conclude that the zero sets of $\Delta_{a,a'}$ are smooth hypersurfaces. Condition (iii) ensures that the regular value theorem is applied in $\text{int}(\mathcal{S})$, which is an open subset of $\mathbb{R}^{d_s}$ (a manifold without boundary); without it, the boundary of $\mathcal{S}$ would require a separate treatment.

Proposition 2.3 (Geometry of the indifference set). *Under Assumption 1, for each pair (a, a') with $\mathcal{K}_{a,a'}(\theta_0) \neq \emptyset$, the set $\mathcal{K}_{a,a'}(\theta_0)$ is a C^1 submanifold of $\text{int}(\mathcal{S})$ with dimension $d_s - 1$. In particular, it has Lebesgue measure zero in $\mathbb{R}^{d_s}$. Since $\mathcal{A}$ is finite, $\mathcal{K}(\theta_0) = \bigcup_{a \neq a'} \mathcal{K}_{a,a'}(\theta_0)$ is a finite union of such manifolds, hence also has Lebesgue measure zero.*

Proof. For each pair (a, a') , the set $\mathcal{K}_{a,a'}(\theta_0)$ is the zero set of the C^1 function $s \mapsto \Delta_{a,a'}(s; \theta_0)$. By Assumption 1(iii), $\mathcal{K}_{a,a'}(\theta_0) \subset \text{int}(\mathcal{S})$, which is an open subset of $\mathbb{R}^{d_s}$ (a manifold without boundary). By Assumption 1(ii), the gradient is nonvanishing at every zero, so by the regular value theorem (Guillemin and Pollack, 1974, Theorem 5.8), the zero set is a C^1 submanifold of codimension one in $\text{int}(\mathcal{S})$. A C^1 submanifold of dimension $d_s - 1$ in $\mathbb{R}^{d_s}$ has Lebesgue measure zero (Guillemin and Pollack, 1974, Theorem 6.3). The union of finitely many measure-zero sets is measure-zero. $\square$

To link this geometric result to probabilistic statements about the stationary distribution, we impose absolute continuity of μ_{θ_0} .

Assumption 2 (Absolutely continuous invariant distribution). *The invariant distribution μ_{θ_0} of the state process $\{s_t\}$ under the true parameter θ_0 admits a bounded density m_0 with respect to Lebesgue measure λ on $\mathcal{S}$:*

$$\mu_{\theta_0}(B) = \int_B m_0(s) d\lambda(s), \quad \sup_{s \in \mathcal{S}} m_0(s) < \infty,$$

for all Borel sets $B \subset \mathcal{S}$.

Corollary 2.4 (Measure-zero indifference). *Under Assumptions 1 and 2, $\mu_{\theta_0}(\mathcal{K}(\theta_0)) = 0$.*

Proof. By Proposition 2.3, $\mathcal{K}(\theta_0)$ has Lebesgue measure zero. Since $\mu_{\theta_0} \ll \lambda$ by Assumption 2, it follows that $\mu_{\theta_0}(\mathcal{K}(\theta_0)) = 0$. $\square$

Remark 2.5 (When absolute continuity fails). *The measure-zero result requires $\mu_{\theta_0} \ll \lambda$. This condition is satisfied in most continuous-state DDC models with additive shocks, since the shock density induces absolute continuity of the transition kernel, which propagates to the invariant distribution under standard mixing conditions (Meyn and Tweedie, 2009). The condition fails for models with discrete state spaces or point-mass components in the invariant distribution. Our theoretical results are therefore restricted to continuous-state DDC models.*

Discussion of the single-crossing condition. In the original version of this paper, we additionally imposed a single-crossing condition requiring that $s \mapsto \Delta_\theta(s; V_\theta)$ be strictly monotonic for binary-action models. While single-crossing simplifies the geometry (it ensures $\mathcal{K}(\theta_0)$ consists of a single point in $d_s = 1$, or a single surface in $d_s > 1$), it is more restrictive than needed. In particular, single-crossing rules out:

- *Replacement models* in the spirit of Rust (1987), where replacement may be optimal at both very low and very high mileage states, yielding a non-monotonic policy;
- *(S, s) adjustment policies* arising from fixed adjustment costs (Cooper and Haltiwanger, 2006), where the inaction region is bounded by two thresholds;
- *Models with multiple indifference surfaces*, as arise in multi-action settings ($J \geq 2$) where different action pairs become indifferent at different state values.

Accordingly, we do *not* impose single-crossing. The transversality condition (Assumption 1) is sufficient: it allows $\mathcal{K}(\theta)$ to consist of *multiple* disjoint $(d_s - 1)$ -dimensional manifolds, provided the gradient of $\Delta_{a,a'}$ is nonvanishing at each.

An illustrative labour-supply example. To make the geometry concrete, consider the dynamic labour-leisure model that motivates our Monte Carlo design. Let $s_t \in \mathcal{S} \subset \mathbb{R}_+$ denote liquid assets and $a_t \in \{0, 1\}$ indicate leisure ($a_t = 0$) or work ($a_t = 1$). Preferences and transitions are

$$u(s, a; \theta) = \frac{(s + wa)^{1-\gamma}}{1-\gamma} - \alpha a, \quad s_{t+1} = \rho s_t + w a_t + \varepsilon_{t+1},$$

with $\theta = (\alpha, \gamma)$, $\alpha > 0$, $\gamma > 0$, wage $w > 0$, $\rho \in (0, 1)$, and i.i.d. shocks ε_{t+1} with a continuous density. Under a natural monotonicity condition on the value-function difference $\Delta_\theta(s; V_\theta) \equiv Q_\theta(s, 1; V_\theta) - Q_\theta(s, 0; V_\theta)$, there exists a unique threshold $\bar{s}(\theta)$ at which $\Delta_\theta(\bar{s}(\theta); V_\theta) = 0$, and the optimal policy is a one-sided threshold rule: $a_\theta^*(s) = 0$ for $s < \bar{s}(\theta)$ and $a_\theta^*(s) = 1$ for $s > \bar{s}(\theta)$. The indifference set is the single point $\mathcal{K}(\theta) = \{\bar{s}(\theta)\}$.

This example illustrates two features. First, the hard-max operator produces a kink in V_θ exactly at the participation threshold: the value function is smooth on either side of $\bar{s}(\theta)$ but non-differentiable at the switching point. Second, the “thinness” of the kink set is transparent: a single point has zero length in $d_s = 1$; in higher dimensions, transversality implies that indifference conditions pin down $(d_s - 1)$ -dimensional surfaces with zero d_s -dimensional volume. Economically, the threshold locates where behaviour changes; geometrically, it is too small to matter for integrals such as likelihoods and moments. This intuition is formalised by Proposition 2.3 and Corollary 2.4.

2.3 The Hard-Max Likelihood

We observe data $\{z_i\}_{i=1}^n = \{(s_i, a_i)\}_{i=1}^n$ from the controlled Markov process under the true parameter θ_0 .

Under the hard-max model, the conditional probability of observing action a given state s is

$$p_{\theta,0}(a|s) = \begin{cases} 1 & \text{if } a \in \Gamma_\theta(s) \text{ and } |\Gamma_\theta(s)| = 1, \\ |\Gamma_\theta(s)|^{-1} & \text{if } a \in \Gamma_\theta(s) \text{ and } |\Gamma_\theta(s)| \geq 2, \\ 0 & \text{if } a \notin \Gamma_\theta(s). \end{cases} \quad (6)$$

Because $p_{\theta,0}(a|s) \in \{0, 1\}$ for μ_{θ_0} -almost every s (by Corollary 2.4), the score $\nabla_\theta \log p_{\theta,0}(a|s)$ does not admit a pointwise expression: the derivative of the indicator function vanishes almost everywhere and is undefined on $\mathcal{K}(\theta)$. Moreover, the log-likelihood $\log p_{\theta,0}(a|s)$ is non-smooth in θ for any fixed (s, a) , because $\Gamma_\theta(s)$ changes discretely when θ crosses values that shift the argmax.

Population criterion. The population log-likelihood is

$$M(\theta) := \mathbb{E}_{P_0}[\log p_{\theta,0}(A|S)]. \quad (7)$$

Under correct specification, $M(\theta_0) = 0$ and $M(\theta) = -\infty$ for $\theta \neq \theta_0$ such that $\{s : a^*(s; \theta_0) \neq a^*(s; \theta)\}$ has positive μ_{θ_0} -measure.

Degeneracy of the hard-max criterion. The hard-max population criterion $M(\theta)$ is degenerate: it equals zero at θ_0 and $-\infty$ elsewhere. While θ_0 is trivially identified as the unique maximiser, the degeneracy prevents standard notions of differentiability from applying to $M(\theta)$ directly. At the population level, the MZI property (Corollary 2.4) implies that the non-smoothness of the Bellman operator and the policy is confined to the set $\mathcal{K}(\theta_0)$, which has probability zero under μ_{θ_0} ; this is the key observation exploited in Section 3.

We address the degeneracy by defining the Hessian and score variance as *limits of the annealed quantities*: $H_0 := \lim_{\tau \downarrow 0} \nabla_{\theta\theta}^2 M(\theta_0, \tau)$ and $\Omega := \lim_{\tau \downarrow 0} \text{Var}(\psi_\tau(Z; \theta_0))$, where $M(\theta, \tau)$ is the annealed population criterion (Definition 3.7).

Sample criterion: the $-\infty$ problem. The sample log-likelihood takes the value $-\infty$ whenever the model at parameter θ predicts a different action than what is observed at *any* data point. This makes the sample criterion extremely non-smooth. Three approaches circumvent this: permanent smoothing (Rust, 1987), constrained optimisation (Su and Judd, 2012), and vanishing smoothing (ANE, this paper).

2.4 Identification

Assumption 3 (Identification). *The policy mapping $\theta \mapsto a^*(\cdot; \theta)$ is injective μ_{θ_0} -a.e.: for every $\theta \neq \theta_0$ in Θ ,*

$$\mu_{\theta_0}(\{s \in \mathcal{S} : a^*(s; \theta) \neq a^*(s; \theta_0)\}) > 0.$$

Remark 2.6. *This identification condition is equivalent to the statement that θ_0 is the unique maximiser of $M(\theta)$ in the extended-real sense, and also equivalent to uniqueness of $\arg \max_\theta M(\theta, \tau)$ for all τ sufficiently small (by Proposition 4.7 below). We do not attempt to derive it from primitive conditions on exclusion restrictions or functional forms (as in Magnac and Thesmar, 2002); instead, we treat it as a high-level assumption that holds under standard conditions in many applied DDC specifications.*

2.5 Data and Stationarity

Assumption 4 (Data). *The data $\{z_i\}_{i=1}^n = \{(s_i, a_i)\}_{i=1}^n$ satisfy one of the following:*

- (a) **Cross-sectional (i.i.d.):** $(z_1, \dots, z_n)$ are i.i.d. draws from $P_0 = \mu_{\theta_0} \otimes p_{\theta_0,0}(\cdot|\cdot)$.
- (b) **Time series (geometrically ergodic):** $\{(s_t, a_t)\}_{t=1}^n$ is a single realisation of the controlled Markov chain under θ_0 , which is geometrically ergodic: there exist $C > 0$ and $\rho \in (0, 1)$ such that $\sup_s \|P^t(\cdot|s) - \mu_{\theta_0}(\cdot)\|_{TV} \leq C\rho^t$ for all $t \geq 1$.

2.6 Formal Assumptions

We collect the regularity conditions on the primitives.

Assumption 5 (Regularity).

- (i) **State space:** $\mathcal{S} \subset \mathbb{R}^{d_s}$ is compact and convex with nonempty interior.
- (ii) **Action space:** $\mathcal{A} = \{0, 1, \dots, J\}$ is finite, with $J \geq 1$.
- (iii) **Parameter space:** $\Theta \subset \mathbb{R}^{d_\theta}$ is compact with $\theta_0 \in \text{int}(\Theta)$.
- (iv) **Utility:** $u : \mathcal{S} \times \mathcal{A} \times \Theta \rightarrow \mathbb{R}$ is bounded, continuous in (s, θ) for each a , and three times continuously differentiable in θ for each (s, a) .
- (v) **Transition kernel:** $F(\cdot|s, a)$ admits a density $f(s'|s, a)$ with respect to Lebesgue measure on $\mathcal{S}$ that is: bounded ($\sup_{s,a,s'} f(s'|s, a) \leq \bar{f} < \infty$); smooth in the current state ($s \mapsto f(s'|s, a)$ is C^1 for each (s', a)); and bounded away from zero on $\text{int}(\mathcal{S})$: for every compact $K \subset \text{int}(\mathcal{S})$, there exists $\underline{f}_K > 0$ such that $f(s'|s, a) \geq \underline{f}_K$ for all $s' \in K$, $(s, a) \in \mathcal{S} \times \mathcal{A}$.
- (vi) **Discount factor:** $\beta \in [0, 1)$.

Assumption 5 is standard in continuous-state DDC models (e.g. Rust, 1987; Magnac and Thesmar, 2002; Aguirregabiria and Mira, 2002). Compactness and boundedness ensure that the Bellman operator is well-defined on $B(\mathcal{S})$, while continuity and differentiability in θ permit differentiation under the integral sign. The bounded density condition guarantees that the transition preserves absolute continuity and that invariant distributions admit densities under standard mixing conditions (Meyn and Tweedie, 2009). The three-times differentiability in condition (iv) is needed for the second-order Taylor expansion in the asymptotic normality proof (Section 6).

Remark 2.7 (Boundary effects from truncation). *In practice, the state space is often truncated to a compact set. Two approaches handle this: (a) choose boundaries sufficiently extreme that the stationary probability of boundary states is negligible, or (b) replace truncation with a smooth reflection scheme. In our Monte Carlo experiments we adopt approach (a).*

3 The Measure-Zero Indifference Property

This section establishes that the Hessian and score variance of the annealed population criterion converge to well-defined, finite limits as $\tau \downarrow 0$, despite the pointwise non-smoothness of the Bellman operator. The central mechanism is geometric: the indifference set $\mathcal{K}(\theta_0)$ where the value function has kinks forms a lower-dimensional manifold with zero probability under μ_{θ_0} (Corollary 2.4), so integrated criteria inherit smoothness from their integrands.

We proceed in three steps. First, we establish Lipschitz continuity and Hadamard directional differentiability of the value function mapping $\theta \mapsto V_\theta$ (Section 3.1–3.2). Second, we develop the MZI lemma showing that integrated criteria are differentiable at θ_0 despite pointwise singularities on $\mathcal{K}(\theta_0)$ (Section 3.3). Third, we define the limiting Hessian H_0 and score variance Ω as limits of annealed quantities and establish their existence (Section 3.4).

3.1 Lipschitz Continuity of the Value Function

We begin with a standard result that will be used repeatedly in the dominated convergence arguments below.

Lemma 3.1 (Lipschitz continuity of V_θ in θ). *Under Assumption 5, there exists $L_V < \infty$ such that for all $\theta, \theta' \in \Theta$,*

$$\|V_\theta - V_{\theta'}\|_\infty \leq L_V \|\theta - \theta'\|. \quad (8)$$

Explicitly, $L_V = C_u/(1 - \beta)$, where $C_u := \sup_{s,a,\theta} \|\nabla_\theta u(s, a; \theta)\| < \infty$.

Proof. For any $V \in B(\mathcal{S})$ and $\theta, \theta' \in \Theta$,

$$|(T_\theta V)(s) - (T_{\theta'} V)(s)| \leq \max_{a \in \mathcal{A}} |u(s, a; \theta) - u(s, a; \theta')| \leq C_u \|\theta - \theta'\|.$$

Since $V_\theta = T_\theta V_\theta$ and $V_{\theta'} = T_{\theta'} V_{\theta'}$,

$$\begin{aligned} \|V_\theta - V_{\theta'}\|_\infty &\leq \|T_\theta V_\theta - T_\theta V_{\theta'}\|_\infty + \|T_\theta V_{\theta'} - T_{\theta'} V_{\theta'}\|_\infty \\ &\leq \beta \|V_\theta - V_{\theta'}\|_\infty + C_u \|\theta - \theta'\|. \end{aligned}$$

Rearranging gives $\|V_\theta - V_{\theta'}\|_\infty \leq C_u \|\theta - \theta'\|/(1 - \beta)$. $\square$

3.2 Directional Differentiability of the Value Function

The value function V_θ depends on θ through the Bellman fixed-point equation $V_\theta = T_\theta V_\theta$. Because T_θ involves a max operator, V_θ is generally not Fréchet differentiable in θ . However, it admits a Hadamard directional derivative.

Proposition 3.2 (Directional Bellman derivative). *Under Assumptions 5–1, for every direction $h \in \mathbb{R}^{d_\theta}$, the directional derivative*

$$W_h(s) := \lim_{t \downarrow 0} \frac{V_{\theta_0 + th}(s) - V_{\theta_0}(s)}{t} \quad (9)$$

exists for all $s \in \mathcal{S}$ in the $\|\cdot\|_\infty$ topology on $B(\mathcal{S})$. Moreover, W_h satisfies the linearised Bellman equation: for μ_{θ_0} -a.e. $s \in \mathcal{S} \setminus \mathcal{K}(\theta_0)$,

$$W_h(s) = \nabla_\theta u(s, a^*(s; \theta_0); \theta_0)^\top h + \beta \int_{\mathcal{S}} W_h(s') F(ds' | s, a^*(s; \theta_0)). \quad (10)$$

On $\mathcal{K}(\theta_0)$, W_h satisfies

$$W_h(s) = \max_{a \in \Gamma_{\theta_0}(s)} \left[\nabla_\theta u(s, a; \theta_0)^\top h + \beta \int_{\mathcal{S}} W_h(s') F(ds' | s, a) \right]. \quad (11)$$

Proof. We apply Bonnans and Shapiro (2000, Theorem 6.3.2) in the Banach space $(B(\mathcal{S}), \|\cdot\|_\infty)$.

Step 1: Hadamard directional differentiability of $\theta \mapsto T_\theta V$ in $\|\cdot\|_\infty$. Fix $V \in B(\mathcal{S})$. For each $s \in \mathcal{S}$, $(T_\theta V)(s) = \max_{a \in \mathcal{A}} Q_\theta(s, a; V)$, where each $\theta \mapsto Q_\theta(s, a; V) = u(s, a; \theta) + \beta \int V(s') F(ds'|s, a)$ is differentiable in θ (Assumption 5(iv)). By the Hadamard directional differentiability of the finite-dimensional max (Bonnans and Shapiro, 2000, Proposition 2.47; see also Appendix C.1), the mapping $\theta \mapsto (T_\theta V)(s)$ is Hadamard directionally differentiable at θ_0 with

$$D_\theta(T_{\theta_0} V)(s)[h] = \max_{a \in \Gamma_{\theta_0, V}(s)} \nabla_\theta u(s, a; \theta_0)^\top h, \quad (12)$$

where $\Gamma_{\theta_0, V}(s) = \arg \max_a Q_{\theta_0}(s, a; V)$.

Step 2: Uniform convergence of the difference quotient. We verify that the convergence in (12) is uniform in $s \in \mathcal{S}$. Since $\mathcal{A}$ is finite and each $\theta \mapsto Q_\theta(s, a; V)$ is differentiable with derivative $\nabla_\theta u(s, a; \theta_0)^\top h$ that is bounded uniformly in s (by Assumption 5(iv)), we have for each a :

$$\sup_{s \in \mathcal{S}} \left| \frac{Q_{\theta_0+th}(s, a; V) - Q_{\theta_0}(s, a; V)}{t} - \nabla_\theta u(s, a; \theta_0)^\top h \right| \rightarrow 0 \quad \text{as } t \rightarrow 0,$$

by the uniform differentiability of u in θ (the derivative $\nabla_\theta u$ is continuous on the compact set $\mathcal{S} \times \mathcal{A} \times \Theta$, hence uniformly continuous). The max of finitely many uniformly converging sequences converges uniformly, so

$$\left\| \frac{T_{\theta_0+th} V - T_{\theta_0} V}{t} - D_\theta(T_{\theta_0} V)[\cdot][h] \right\|_\infty \rightarrow 0.$$

Step 3: Contraction property. T_θ is a contraction on $(B(\mathcal{S}), \|\cdot\|_\infty)$ with modulus $\beta < 1$ for each θ , uniformly in $\theta \in \Theta$.

Step 4: Application of Bonnans-Shapiro. Steps 1–3 verify the hypotheses of Bonnans and Shapiro (2000, Theorem 6.3.2). The conclusion is that the fixed point V_θ is Hadamard directionally differentiable at θ_0 in $(B(\mathcal{S}), \|\cdot\|_\infty)$. The directional derivative W_h is the unique solution of the linearised fixed-point equation

$$W_h(s) = D_\theta(T_{\theta_0} V_{\theta_0})(s)[h] + D_V(T_{\theta_0} V_{\theta_0})(s)[W_h],$$

where $D_V(T_{\theta_0} V_{\theta_0})(s)[W] = \beta \int W(s') F(ds'|s, a^*(s; \theta_0))$ for $s \notin \mathcal{K}(\theta_0)$. Using (12), this yields (10) on $\mathcal{S} \setminus \mathcal{K}(\theta_0)$ and (11) on $\mathcal{K}(\theta_0)$.

Step 5: Linearity and convergence topology. Since $D_\theta(T_{\theta_0} V)(s)[h] = \max_{a \in \Gamma(s)} \nabla_\theta u(s, a; \theta_0)^\top h$ is positively homogeneous (but not linear) in h on $\mathcal{K}(\theta_0)$, W_h is Hadamard (but not Gateaux) directionally differentiable in h at points $s \in \mathcal{K}(\theta_0)$. On $\mathcal{S} \setminus \mathcal{K}(\theta_0)$, the max reduces to the unique optimal action, and W_h is linear in h . Since $\mu_{\theta_0}(\mathcal{K}(\theta_0)) = 0$, W_h depends linearly on h in $L^p(\mu_{\theta_0})$ for any $p \geq 1$. $\square$

Remark 3.3 (Why Hadamard, not Fréchet). *The convergence in (9) holds in $\|\cdot\|_\infty$ (not merely L^1), which is stronger than $L^1(\mu_{\theta_0})$ convergence. However, V_θ is Hadamard directionally differentiable but not Fréchet differentiable, because the directional derivative W_h is not linear in h on $\mathcal{K}(\theta_0)$ (it involves max over the argmax set). For integrated criteria, Hadamard differentiability suffices: the chain rule for Hadamard derivatives combined with dominated convergence yields the correct derivative formula (Lemma 3.5 below).*

Remark 3.4 (Comparison with standard DDC approaches). *Many papers in the DDC literature assume V_θ is everywhere differentiable in θ and apply the standard chain rule. This is incorrect*

when V_θ has kinks at $\mathcal{K}(\theta)$. Proposition 3.2 handles this rigorously by: (i) working in $\|\cdot\|_\infty$ via the Bonnans-Shapiro implicit function theorem for parameterised contractions, which provides the correct derivative formula even when the maximiser in T_θ is not unique everywhere; (ii) obtaining only Hadamard (not Fréchet) differentiability, which is sufficient for integrated criteria since the non-linear part is confined to $\mathcal{K}(\theta_0)$.

3.3 The MZI Lemma

The following result shows that integrated criteria depending on (V_θ, θ) are differentiable at θ_0 despite pointwise non-smoothness, provided the singularities occur on a set of μ_{θ_0} -measure zero.

Lemma 3.5 (Measure-zero indifference). *Under Assumptions 5–3, let $g : \mathcal{S} \times \mathcal{A} \times \Theta \rightarrow \mathbb{R}$ be measurable and suppose that, for each $s \notin \mathcal{K}(\theta_0)$ and each $a \in \mathcal{A}$:*

- (i) $\theta \mapsto g(s, a, \theta)$ is differentiable at θ_0 when V_θ is held fixed at V_{θ_0} ;
- (ii) $V \mapsto g(s, a, \theta_0; V)$ is Fréchet differentiable at V_{θ_0} in $(B(\mathcal{S}), \|\cdot\|_\infty)$, with derivative denoted $D_V g(s, a, \theta_0)[\cdot]$;
- (iii) there exists an integrable function $\bar{g}(s, a)$ such that $|g(s, a, \theta) - g(s, a, \theta_0)| \leq \bar{g}(s, a) \cdot \|\theta - \theta_0\|$ for all θ in a neighbourhood of θ_0 .

Then the integrated criterion $G(\theta) := \mathbb{E}_{P_0}[g(S, A, \theta; V_\theta)]$ is differentiable at θ_0 , and

$$\nabla_\theta G(\theta_0) = \mathbb{E}_{P_0}[\nabla_\theta g(S, A, \theta_0; V_{\theta_0}) + D_V g(S, A, \theta_0)[W_h]], \quad (13)$$

where W_h is the directional derivative of V_θ from Proposition 3.2 and $\nabla_\theta g$ denotes the partial derivative holding V fixed.

Proof. Fix $h \in \mathbb{R}^{d_\theta}$ and let $\theta_t = \theta_0 + th$. Write

$$\frac{G(\theta_t) - G(\theta_0)}{t} = \underbrace{\mathbb{E}_{P_0}\left[\frac{g(S, A, \theta_t; V_{\theta_t}) - g(S, A, \theta_0; V_{\theta_0})}{t}\right]}_{=: I_1(t)} + \underbrace{\mathbb{E}_{P_0}\left[\frac{g(S, A, \theta_t; V_{\theta_t}) - g(S, A, \theta_t; V_{\theta_0})}{t}\right]}_{=: I_2(t)}.$$

Term I_1 (direct effect). For $s \notin \mathcal{K}(\theta_0)$, condition (i) gives pointwise convergence of the difference quotient. The envelope $\bar{g}$ from condition (iii) provides an integrable dominating function. Since $\mu_{\theta_0}(\mathcal{K}(\theta_0)) = 0$ (Corollary 2.4), dominated convergence gives $I_1(t) \rightarrow \mathbb{E}_{P_0}[\nabla_\theta g(S, A, \theta_0; V_{\theta_0})^\top h]$.

Term I_2 (indirect effect through V_θ). By condition (ii) and the definition of Fréchet differentiability:

$$g(s, a, \theta_t; V_{\theta_t}) - g(s, a, \theta_t; V_{\theta_0}) = D_V g(s, a, \theta_0)[V_{\theta_t} - V_{\theta_0}] + o(\|V_{\theta_t} - V_{\theta_0}\|_\infty).$$

Dividing by t and using $t^{-1}(V_{\theta_t} - V_{\theta_0}) \rightarrow W_h$ in $\|\cdot\|_\infty$ (Proposition 3.2), together with the continuity of $D_V g$ as a bounded linear functional on $(B(\mathcal{S}), \|\cdot\|_\infty)$, gives $I_2(t) \rightarrow \mathbb{E}_{P_0}[D_V g(S, A, \theta_0)[W_h]]$.

The remainder term is controlled by $|o(\|V_{\theta_t} - V_{\theta_0}\|_\infty)|/t = o(L_V \|h\|) \rightarrow 0$ (using Lemma 3.1), and the dominated convergence is justified by condition (iii) together with $\|V_{\theta_t} - V_{\theta_0}\|_\infty \leq L_V t \|h\|$.

Since this holds for all h and W_h depends linearly on h in $L^1(\mu_{\theta_0})$, G is differentiable at θ_0 . $\square$

Remark 3.6 (Role of the envelope and Lipschitz conditions). *Condition (iii) in Lemma 3.5 and Lemma 3.1 together ensure that the remainder term in I_2 vanishes after normalisation by t . The key chain is: $\|V_{\theta_t} - V_{\theta_0}\|_\infty \leq L_V t \|h\|$ (Lemma 3.1), so the Fréchet remainder satisfies $|r_t(s, a)|/t \leq \bar{g}(s, a) \cdot (1 + L_V) \|h\|$, which is integrable. The Lipschitz bound thus converts the $\|\cdot\|_\infty$ convergence of V_{θ_t} into the integrability bound needed for dominated convergence.*

3.4 Smoothness of the Population Criterion via Annealed Limits

As noted in Section 2.3, the hard-max population criterion $M(\theta)$ is degenerate ($M(\theta_0) = 0$, $M(\theta) = -\infty$ for $\theta \neq \theta_0$). The relevant objects for the asymptotic theory are the Hessian and score variance of the *annealed* criterion, in the limit as $\tau \downarrow 0$.

Definition 3.7 (Annealed Hessian and score variance). *Define the annealed population criterion $M(\theta, \tau) := \mathbb{E}_{P_0}[\ell(Z; \theta, \tau; V_{\theta, \tau})]$ (see Section 4). The limiting Hessian and limiting score variance are*

$$H_0 := \lim_{\tau \downarrow 0} \nabla_{\theta\theta}^2 M(\theta_0, \tau), \quad (14)$$

$$\Omega := \lim_{\tau \downarrow 0} \text{Var}_{P_0}(\psi_\tau(Z; \theta_0)), \quad (15)$$

provided these limits exist.

Theorem 3.8 (Existence and regularity of limiting quantities). *Under Assumptions 5–3 and the margin condition (22) with exponent $\kappa \geq 2$:*

- (i) *The limit H_0 in (14) exists and is finite. Under Assumption 3, H_0 is negative semidefinite.*
- (ii) *The limit Ω in (15) exists and is finite.*
- (iii) *$\nabla_{\theta} M(\theta_0, \tau) = \mathbb{E}_{P_0}[\psi_\tau(Z; \theta_0)] \rightarrow 0$ as $\tau \downarrow 0$, consistently with the first-order condition of the true model.*

Under the stronger condition $\kappa > 2$, Ω is strictly positive definite (assuming Assumption 8(i) below).

Proof. Part (i). By Proposition 4.4 (see Section 4), $|\nabla^2 M(\theta_0, \tau) - \nabla^2 M(\theta_0, \tau')| \leq C(|\tau|^\kappa + |\tau'|^\kappa)[\log(1/\min(\tau, \tau'))]^\kappa$ for any $0 < \tau' \leq \tau \leq 1$. For $\kappa \geq 2$, this is a Cauchy sequence in τ , so the limit H_0 exists and is finite. Negative semidefiniteness: since θ_0 maximises $M(\theta, \tau)$ for all τ sufficiently small (Proposition 4.7), $\nabla^2 M(\theta_0, \tau) \preceq 0$ for all such τ , and the limit inherits this.

Part (ii). Write $\Omega(\tau) = \mathbb{E}[\psi_\tau \psi_\tau^\top] - \mathbb{E}[\psi_\tau] \mathbb{E}[\psi_\tau]^\top$. By the explicit computation in Appendix C.5 (Example B.3), $\mathbb{E}[\|\psi_\tau\|^2] = O(\tau^{\kappa-2})$ under the margin condition. For $\kappa = 2$, $\mathbb{E}[\|\psi_\tau\|^2]$ converges to a finite constant. For $\kappa > 2$, $\mathbb{E}[\|\psi_\tau\|^2] \rightarrow 0$; in this case $\Omega = 0$ and the CLT gives a faster rate than $\sqrt{n}$ (Remark 3.10).

To show convergence (not just boundedness), we verify that $\tau \mapsto \mathbb{E}[\psi_\tau \psi_\tau^\top]$ is continuous from the right at $\tau = 0$, using dominated convergence on the decomposition into near-threshold ($\{\Delta \leq \delta\}$) and far-from-threshold ($\{\Delta > \delta\}$) regions; see Appendix C.5.

Part (iii). $\nabla_{\theta} M(\theta_0, \tau) = \mathbb{E}[\psi_\tau(Z; \theta_0)]$ is the mean score at the true parameter. By Proposition 4.4, $\|\nabla_{\theta} M(\theta_0, \tau)\| \leq C' \tau^\kappa [\log(1/\tau)]^\kappa \rightarrow 0$ as $\tau \downarrow 0$. $\square$

Remark 3.9 (Role of the margin exponent κ). *The margin exponent κ governs the probability mass near the indifference boundary. When $\kappa < 2$, the score variance $\text{Var}(\psi_\tau)$ diverges as $\tau \rightarrow 0$. In this case, the standard $\sqrt{n}$ normalisation is no longer appropriate, and a different rate and limiting distribution may arise. We restrict the asymptotic theory to $\kappa \geq 2$, which requires that the invariant density vanishes at the threshold (or at least has a thin tail near it). This is satisfied in models where the transition density has bounded support and the threshold is away from the mode of the invariant distribution. The Monte Carlo model in Section 8 has $\kappa \approx 1$ (approximately normal density near the threshold), which places it outside the formal theory; we discuss the implications in Section 8.5.*

Remark 3.10 (Faster rate when $\kappa > 2$). When $\kappa > 2$, $\Omega = 0$ and the CLT as stated gives a degenerate limit. The correct scaling is then faster than $\sqrt{n}$: by choosing τ_n to balance the bias and variance terms, one can show that $n^{\tau\kappa}(\hat{\theta}_n - \theta_0)$ has a non-degenerate limit for $\tau\kappa > 1/2$ depending on κ . We do not pursue this extension here.

Remark 3.11 (Rate implications). Theorem 3.8 implies that the annealed population criterion $M(\theta, \tau)$ behaves locally like a smooth quadratic in θ near θ_0 for each small τ , and the curvature converges to H_0 as $\tau \rightarrow 0$. As a result, estimators that maximise a consistent approximation to $M(\theta, \tau_n)$ with $\tau_n \rightarrow 0$ inherit standard $\sqrt{n}$ -rate asymptotics and Gaussian limits (Section 6), rather than the cube-root behaviour characteristic of non-smooth M -estimators such as maximum score (Kim and Pollard, 1990). The key distinction is that the criterion is smooth in expectation (because singularities occur on a μ_{θ_0} -null set), even though it is non-smooth pointwise.

Remark 3.12 (Efficiency). While Theorem 3.8 establishes that H_0 and Ω are well-defined, we have not proven that the asymptotic variance $H_0^{-1}\Omega H_0^{-1}$ achieves the semiparametric efficiency bound for estimating θ_0 in the hard-max DDC model. Establishing efficiency would require deriving the tangent space and verifying that the influence function lies in it. Whether the ANE estimator achieves the efficiency bound remains an open question.

4 Annealed Estimation

This section embeds the hard-max dynamic program in a family of smooth, log-sum-exp Bellman operators indexed by a temperature parameter $\tau > 0$. The annealed operators serve purely as numerical devices: for each fixed $\tau > 0$ they regularise the optimisation problem and yield smooth conditional choice probabilities, while as $\tau \rightarrow 0$ they converge uniformly to the hard-max operator.

4.1 The Annealed Bellman Operator

For temperature $\tau > 0$, the *annealed* (soft-max) Bellman operator replaces the hard max with the log-sum-exp smooth approximation:

$$(T_\theta^\tau V)(s) := \tau \log \sum_{a \in \mathcal{A}} \exp \left[\frac{Q_\theta(s, a; V)}{\tau} \right]. \quad (16)$$

The log-sum-exp function satisfies the basic sandwich inequality

$$\max_{a \in \mathcal{A}} x_a \leq \tau \log \sum_{a \in \mathcal{A}} \exp(x_a/\tau) \leq \max_{a \in \mathcal{A}} x_a + \tau \log |\mathcal{A}| \quad (17)$$

for any $x \in \mathbb{R}^{|\mathcal{A}|}$, with convergence to $\max_a x_a$ as $\tau \downarrow 0$. Furthermore, the log-sum-exp is a contraction in the ℓ^∞ sense:

$$\left| \tau \log \sum_a \exp(x_a/\tau) - \tau \log \sum_a \exp(y_a/\tau) \right| \leq \max_a |x_a - y_a|, \quad (18)$$

which follows from rewriting the ratio using softmax weights and applying Jensen's inequality.

For any $\tau > 0$, T_θ^τ is a contraction on $B(\mathcal{S})$ with modulus β : by (18) and $|Q_\theta(s, a; V) - Q_\theta(s, a; V')| \leq \beta \|V - V'\|_\infty$ (from (2)),

$$\|T_\theta^\tau V - T_\theta^\tau V'\|_\infty \leq \beta \|V - V'\|_\infty.$$

The Banach fixed-point theorem gives a unique fixed point $V_{\theta,\tau} \in B(\mathcal{S})$.

The associated annealed conditional choice probability is

$$p_{\theta,\tau}(a|s; V) = \frac{\exp[Q_{\theta}(s, a; V)/\tau]}{\sum_{a'} \exp[Q_{\theta}(s, a'; V)/\tau]}, \quad (19)$$

which is smooth in (θ, V, τ, s) for $\tau > 0$ and has full support on $\mathcal{A}$. These CCPs describe the response of the annealed Bellman operator to the value function; they are *not* part of the behavioural model, which remains hard-max. The temperature τ is a computational device, not a structural parameter.

4.2 Convergence as $\tau \rightarrow 0$

Lemma 4.1 (Convergence of annealed value functions). *Under Assumption 5,*

$$\|V_{\theta,\tau} - V_{\theta}\|_{\infty} \leq \frac{\tau \log |\mathcal{A}|}{1 - \beta} \rightarrow 0 \quad \text{as } \tau \downarrow 0, \quad (20)$$

uniformly in $\theta \in \Theta$.

Proof. By (17), $\|T_{\theta}^{\tau}V - T_{\theta}V\|_{\infty} \leq \tau \log |\mathcal{A}|$ for any $V \in B(\mathcal{S})$. Since both T_{θ}^{τ} and T_{θ} are β -contractions with fixed points $V_{\theta,\tau}$ and V_{θ} respectively,

$$\|V_{\theta,\tau} - V_{\theta}\|_{\infty} \leq \beta \|V_{\theta,\tau} - V_{\theta}\|_{\infty} + \tau \log |\mathcal{A}|,$$

and rearranging gives (20). $\square$

Lemma 4.2 (Convergence of annealed CCPs). *For every $s \notin \mathcal{K}(\theta)$ and $a \in \mathcal{A}$, $p_{\theta,\tau}(a|s; V_{\theta,\tau}) \rightarrow \mathbf{1}\{a = a^*(s; \theta)\}$ as $\tau \downarrow 0$. The convergence is exponential: for $s \notin \mathcal{K}(\theta)$,*

$$p_{\theta,\tau}(a^*(s; \theta)|s; V_{\theta,\tau}) \geq 1 - |\mathcal{A}| \exp\left\{-\frac{\delta(s; \theta)}{2\tau}\right\}, \quad (21)$$

where $\delta(s; \theta) := Q_{\theta}(s, a^*; V_{\theta}) - \max_{a \neq a^*} Q_{\theta}(s, a; V_{\theta}) > 0$ is the optimality gap.

Proof. At $s \notin \mathcal{K}(\theta)$, the unique optimal action a^* has gap $\delta(s; \theta) > 0$. By Lemma 4.1, $|Q_{\theta}(s, a; V_{\theta,\tau}) - Q_{\theta}(s, a; V_{\theta})| \leq \beta \tau \log |\mathcal{A}|/(1 - \beta)$ for each a , so the gap under $V_{\theta,\tau}$ satisfies $\delta_{\tau}(s; \theta) \geq \delta(s; \theta)/2$ for all τ sufficiently small. Writing the softmax in ratio form,

$$p_{\theta,\tau}(a^*|s) = \frac{1}{1 + \sum_{a \neq a^*} \exp\{[Q_{\theta}(s, a; V_{\theta,\tau}) - Q_{\theta}(s, a^*; V_{\theta,\tau})]/\tau\}} \geq \frac{1}{1 + (|\mathcal{A}| - 1) \exp\{-\delta(s; \theta)/(2\tau)\}},$$

which gives (21) via $1/(1+x) \geq 1-x$. $\square$

Remark 4.3 (Why $O(\tau^{\kappa})$ for the population criterion, not $O(e^{-c/\tau})$). *Although individual CCPs concentrate exponentially (Lemma 4.2), the population likelihood difference $M(\theta, \tau) - M(\theta, 0)$ converges only at rate $O(\tau^{\kappa})$. This is because states near the indifference boundary have gap $\delta(s; \theta) = O(\tau)$, so the exponential bound collapses at these states. When integrating over all states, the contribution from the near-threshold region $\{s : \delta(s; \theta) \leq \delta\}$ —whose μ_{θ_0} -measure is $O(\delta^{\kappa})$ under the margin condition—dominates the rate. The margin exponent κ thus governs the speed at which the population-level bias vanishes.*

Proposition 4.4 (Smoothing bias bound). *Suppose there exist $\kappa > 0$ and $C_{\kappa} > 0$ such that the margin condition holds:*

$$\mu_{\theta_0}(\{s : \min_{a \neq a^*} \Delta_{a^*,a}(s; \theta_0) \leq \delta\}) \leq C_{\kappa} \delta^{\kappa} \quad \forall \delta > 0. \quad (22)$$

Then for all θ in a neighbourhood of θ_0 where a uniform margin condition holds with the same exponent κ , $|M(\theta, \tau) - M(\theta, 0)| \leq C \tau^{\kappa} [\log(1/\tau)]^{\kappa}$ for a constant C depending on $(C_{\kappa}, \beta, \bar{u}, |\mathcal{A}|)$.

Proof. See Appendix C.2. $\square$

4.3 Annealed Likelihood and Score

The annealed individual log-likelihood is

$$\ell(z; \theta, \tau; V) := \log p_{\theta, \tau}(a|s; V) = \frac{Q_{\theta}(s, a; V)}{\tau} - \log \sum_{a'} \exp\left[\frac{Q_{\theta}(s, a'; V)}{\tau}\right]. \quad (23)$$

The annealed score is $\psi_{\tau}(z; \theta) := \nabla_{\theta} \ell(z; \theta, \tau; V_{\theta, \tau})$.

Lemma 4.5 (Lipschitz bound for the annealed log-likelihood in V). *For any $\tau > 0$, $\theta \in \Theta$, and $V_1, V_2 \in B(\mathcal{S})$,*

$$|\ell(z; \theta, \tau; V_1) - \ell(z; \theta, \tau; V_2)| \leq \frac{2\beta}{\tau} \|V_1 - V_2\|_{\infty}. \quad (24)$$

Proof. See Appendix C.4. $\square$

The factor $1/\tau$ in (24) reflects the sensitivity of the softmax to the value function scale and drives the rate condition $\sqrt{n} \varepsilon_n / \tau_n \rightarrow 0$ (Section 5.3): the sieve approximation error ε_n must shrink faster than τ_n to prevent the Lipschitz amplification from contaminating the score.

Lemma 4.6 (Score convergence in L^2). *Under Assumptions 5–1 and the margin condition (22) with $\kappa \geq 2$,*

$$\text{Var}(\psi_{\tau}(Z; \theta_0)) \rightarrow \Omega$$

as $\tau \downarrow 0$, where Ω is finite and equals

$$\Omega = \lim_{\tau \downarrow 0} \mathbb{E}_{P_0} [\psi_{\tau}(Z; \theta_0) \psi_{\tau}(Z; \theta_0)^{\top}]. \quad (25)$$

Proof. See Appendix C.5. $\square$

4.4 Annealing Neutrality

Neutrality means that the annealed criterion preserves the estimation target: for all sufficiently small τ , the maximiser of $M(\theta, \tau)$ is θ_0 , the same as for the hard-max criterion.

Proposition 4.7 (Annealing neutrality). *Under Assumptions 5–3,*

$$\lim_{\tau \downarrow 0} \operatorname{argmax}_{\theta \in \Theta} M(\theta, \tau) = \{\theta_0\}.$$

More precisely, for every $\varepsilon > 0$ there exists $\bar{\tau} > 0$ such that $\operatorname{argmax}_{\theta \in \Theta} M(\theta, \tau) \subset B(\theta_0, \varepsilon)$ for all $\tau \in (0, \bar{\tau}]$.

Proof. Step 1: $M(\theta_0, \tau) \rightarrow 0$ as $\tau \downarrow 0$. At $\theta = \theta_0$, the true action $a^* = a^*(s; \theta_0)$ is observed P_0 -a.s. on $\mathcal{S} \setminus \mathcal{K}(\theta_0)$. For such s , $\log p_{\theta_0, \tau}(a^*|s; V_{\theta_0, \tau}) \rightarrow \log 1 = 0$ as $\tau \rightarrow 0$ (Lemma 4.2). The integrand is bounded: $\log p_{\theta_0, \tau}(a^*|s) \in [-\log |\mathcal{A}|, 0]$. By dominated convergence, $M(\theta_0, \tau) \rightarrow 0$.

Step 2: Misclassification bound for θ away from θ_0 . Fix $\varepsilon > 0$. By Assumption 3, for every $\theta \in \Theta \setminus B(\theta_0, \varepsilon)$:

$$\eta(\varepsilon) := \inf_{\theta \in \Theta \setminus B(\theta_0, \varepsilon)} \mu_{\theta_0}(\{s : a^*(s; \theta) \neq a^*(s; \theta_0)\}) > 0.$$

(The infimum is positive because: (a) identification gives $\mu_{\theta_0}(\cdots) > 0$ for each such θ ; (b) $\Theta \setminus B(\theta_0, \varepsilon)$ is compact; (c) $\theta \mapsto \mu_{\theta_0}(\{s : a^*(s; \theta) \neq a^*(s; \theta_0)\})$ is lower semicontinuous by Fatou's lemma.)

On the set $\{s : a^*(s; \theta) \neq a^*(s; \theta_0)\}$, the observed action $a^*(s; \theta_0)$ is not optimal at θ , so $p_{\theta, \tau}(a^*(s; \theta_0)|s; V_{\theta, \tau}) \leq 1 - p_{\theta, \tau}(a^*(s; \theta)|s; V_{\theta, \tau})$. For τ small enough, Lemma 4.2 ensures $p_{\theta, \tau}(a^*(s; \theta)|s) \geq 1/2$ on most of the misclassified set, giving $\log p_{\theta, \tau}(a^*(s; \theta_0)|s) \leq -\log 2$ on a subset of μ_{θ_0} -measure at least $\eta(\varepsilon)/2$. Therefore:

$$M(\theta, \tau) \leq -\frac{\eta(\varepsilon)}{2} \log 2 =: -c(\varepsilon) < 0$$

for all τ sufficiently small and all $\theta \in \Theta \setminus B(\theta_0, \varepsilon)$.

Step 3: Combining. For $\tau \leq \min(\tau^*(\varepsilon), \bar{\tau}_1)$ where $M(\theta_0, \tau) > -c(\varepsilon)/2$, we have $M(\theta_0, \tau) > -c(\varepsilon)/2 > -c(\varepsilon) \geq M(\theta, \tau)$ for all $\theta \notin B(\theta_0, \varepsilon)$. Hence any maximiser of $M(\cdot, \tau)$ must lie in $B(\theta_0, \varepsilon)$. $\square$

Remark 4.8 (Population versus sample neutrality). *Proposition 4.7 is a population-level result: it states that the annealed population criterion $M(\theta, \tau)$ has the same maximiser as the hard-max criterion $M(\theta, 0)$ for small τ . At the sample level, the ANE estimator $\hat{\theta}_n$ will differ from the infeasible hard-max MLE in finite samples due to smoothing bias ($\tau_n > 0$), sieve approximation error, and sampling variability. Under the rate conditions in Section 5.3, these three sources of error are each $o_p(n^{-1/2})$, so they do not affect the limiting distribution (Section 6).*

5 Sieve Approximation

The annealed Bellman operators in Section 4 provide smooth mappings from value functions to value functions for each (θ, τ) . In high-dimensional state spaces, however, we cannot evaluate $V_{\theta, \tau}$ on a full grid; we must approximate it in a finite-dimensional function class. This section introduces a neural-network sieve for value function approximation, establishes approximation error bounds, and derives the feasible rate conditions that ensure sieve and annealing errors are jointly negligible at the $\sqrt{n}$ scale.

5.1 Neural Network Architecture

We approximate V_θ by a fully connected feedforward neural network with ReLU activations. Each network $v \in \mathcal{V}_n$ has the form

$$v(s) = W_L \sigma(W_{L-1} \sigma(\cdots \sigma(W_1 s + b_1) \cdots) + b_{L-1}) + b_L, \quad (26)$$

where $\sigma(x) = \max\{0, x\}$ is the ReLU activation applied element-wise, W_ℓ are weight matrices, and b_ℓ are bias vectors. The sieve space $\mathcal{V}_n$ has depth $L_n = O(\log n)$, width $K_n = O(n^{d_s/(2r+d_s)})$, and total parameters $p_n = O(n^{d_s/(2r+d_s)} \log n)$.

Given (θ, τ) and the sieve class $\mathcal{V}_n$, define $\hat{V}_{\theta, \tau}^{(n)} \in \mathcal{V}_n$ as an approximate fixed point of the annealed Bellman operator T_θ^τ computed by fitted value iteration:

$$\hat{V}_{\theta, \tau}^{(k+1)} \in \underset{v \in \mathcal{V}_n}{\operatorname{argmin}} \|v - T_\theta^\tau \hat{V}_{\theta, \tau}^{(k)}\|_n, \quad (27)$$

where $\|\cdot\|_n$ denotes an empirical norm (mean squared error over a set of grid points or Monte Carlo draws) and iteration continues until $\|\hat{V}_{\theta, \tau}^{(k+1)} - \hat{V}_{\theta, \tau}^{(k)}\|_n < \delta_n$ for a numerical tolerance $\delta_n \downarrow 0$.

Lemma 5.1 (Contraction-based error bound). *Under Assumption 5, if $\hat{V}_{\theta,\tau}^{(n)}$ satisfies $\|\hat{V}_{\theta,\tau}^{(n)} - T_\theta^\tau \hat{V}_{\theta,\tau}^{(n)}\|_\infty \leq \delta_n$, then*

$$\|\hat{V}_{\theta,\tau}^{(n)} - V_{\theta,\tau}\|_\infty \leq \frac{\delta_n + 2e_n}{1 - \beta}, \quad (28)$$

where $e_n := \inf_{v \in \mathcal{V}_n} \|v - V_{\theta,\tau}\|_\infty$ is the best-in-class approximation error.

Proof. Let $v^* \in \mathcal{V}_n$ attain $\|v^* - V_{\theta,\tau}\|_\infty = e_n$. Using $V_{\theta,\tau} = T_\theta^\tau V_{\theta,\tau}$ and the β -contraction property of T_θ^τ (Lemma 4.1),

$$\begin{aligned} \|\hat{V}_{\theta,\tau}^{(n)} - v^*\|_\infty &\leq \|\hat{V}_{\theta,\tau}^{(n)} - T_\theta^\tau \hat{V}_{\theta,\tau}^{(n)}\|_\infty + \|T_\theta^\tau \hat{V}_{\theta,\tau}^{(n)} - T_\theta^\tau v^*\|_\infty + \|T_\theta^\tau v^* - v^*\|_\infty \\ &\leq \delta_n + \beta \|\hat{V}_{\theta,\tau}^{(n)} - v^*\|_\infty + (\beta + 1)e_n. \end{aligned}$$

Rearranging and adding $e_n = \|v^* - V_{\theta,\tau}\|_\infty$ gives the result. $\square$

5.2 Approximation Rates

Assumption 6 (Smoothness away from the kink). $V_\theta \in C^r(\mathcal{S} \setminus \mathcal{K}(\theta))$ for some $r \geq 2$, uniformly in $\theta \in \Theta$.

Proposition 5.2 (Sieve approximation rate). *Under Assumption 6 with $r \geq 2$,*

$$\varepsilon_n := \sup_{\theta \in \Theta} \|\hat{V}_\theta^{(n)} - V_\theta\|_\infty = O\left(n^{-r/(2r+d_s)} (\log n)^2\right). \quad (29)$$

Proof. The result follows from the deep-ReLU approximation theory of Yarotsky (2017) (see also Schmidt-Hieber 2020): for $f \in C^r([0, 1]^{d_s})$, a ReLU network with N parameters achieves $\|f - f_N\|_\infty = O(N^{-r/d_s})$. With $N_n = O(n^{d_s/(2r+d_s)} \log n)$ parameters, this yields $e_n = O(n^{-r/(2r+d_s)} (\log n)^2)$. Lemma 5.1 then gives ε_n at the same rate (up to the $(1 - \beta)^{-1}$ factor). $\square$

Remark 5.3 (Kinks do not degrade the approximation rate). *Although V_θ has kinks at $\mathcal{K}(\theta)$, these do not degrade the approximation rate in L^∞ because: (i) $\mathcal{K}(\theta)$ is a $(d_s - 1)$ -dimensional hypersurface (Assumption 1); (ii) ReLU networks can represent kinks exactly (being piecewise linear), so the non-smooth locus does not increase the approximation complexity (Farrell et al., 2021, Section 3.3); (iii) the Lipschitz bound from Lemma 3.1 ensures the kink has bounded height, so the approximation error near $\mathcal{K}(\theta)$ is controlled by the global Lipschitz constant $L_V = C_u/(1 - \beta)$. The effective rate is thus determined by the smoothness r on $\mathcal{S} \setminus \mathcal{K}(\theta)$, not by the global regularity.*

5.3 Feasible Rate Conditions

Assumption 7 (Rate conditions). *The annealing schedule $\tau_n = n^{-\beta}$ and the sieve approximation rate $\varepsilon_n = O(n^{-\alpha})$ satisfy:*

$$(R1) \quad \tau_n \rightarrow 0: \quad \beta > 0;$$

$$(R2) \quad \sqrt{n} \tau_n \rightarrow \infty: \quad \beta < 1/2;$$

$$(R3) \quad \sqrt{n} \varepsilon_n / \tau_n \rightarrow 0: \quad \alpha - \beta > 1/2;$$

$$(R4) \quad \sqrt{n} \tau_n^\kappa \rightarrow 0: \quad \kappa\beta > 1/2.$$

Condition (R3) deserves emphasis. The Lipschitz bound $2\beta/\tau$ in Lemma 4.5 amplifies the sieve approximation error ε_n by a factor $1/\tau_n$ when it enters the score. The requirement $\sqrt{n}\varepsilon_n/\tau_n \rightarrow 0$ ensures this amplified error remains negligible at the $\sqrt{n}$ scale. Since $n^{1/2} \cdot n^{-\alpha}/n^{-\beta} = n^{1/2-\alpha+\beta}$, we need $1/2 - \alpha + \beta < 0$, i.e., $\beta < \alpha - 1/2$.

Proposition 5.4 (Existence of feasible rates). *A triple (α, β, κ) satisfying (R1)–(R4) exists if and only if*

$$\alpha > \frac{1}{2} + \frac{1}{2\kappa}. \quad (30)$$

With the neural-network rate $\alpha = r/(2r + d_s)$, this requires $r/(2r + d_s) > 1/2 + 1/(2\kappa)$.

Proof. The constraints on β are: $\beta \in (0, 1/2)$ from (R1)–(R2), $\beta < \alpha - 1/2$ from (R3), and $\beta > 1/(2\kappa)$ from (R4). A feasible β exists iff $1/(2\kappa) < \min(\alpha - 1/2, 1/2)$, which requires $\alpha > 1/2 + 1/(2\kappa)$. $\square$

Remark 5.5 (Feasibility is restrictive with neural-network sieves). *Condition (30) with $\alpha = r/(2r + d_s)$ is demanding. Table 2 reports the minimum κ required for feasibility at various (d_s, r) combinations.*

Table 2: Minimum margin exponent κ for neural-network feasibility

d_s	r	$\alpha = r/(2r + d_s)$	Min κ	Feasible?
1	2	0.400	—	No ($\alpha < 0.5$)
1	3	0.429	—	No ($\alpha < 0.5$)
1	5	0.455	—	No ($\alpha < 0.5$)
1	10	0.476	—	No ($\alpha < 0.5$)
2	3	0.375	—	No
1	∞	0.500	—	Borderline
Grid ($\varepsilon_n = 0$)		—	> 1	Yes

For $d_s = 1$, one needs $\alpha > 1/2$, i.e., $r/(2r + 1) > 1/2$, i.e., $r > 2r + 1$, which is impossible. **No finite smoothness r suffices for the neural-network rate conditions in $d_s = 1$.** The fundamental reason is that the neural-network approximation rate $r/(2r + d_s)$ cannot exceed $1/2$ for any finite r and $d_s \geq 1$.

We draw three conclusions:

1. The **grid-based ANE** (no sieve approximation, $\varepsilon_n = 0$, (R3) vacuous) is the operative theoretical result. It is feasible for any $\kappa > 1$ with $\beta \in (1/(2\kappa), 1/2)$. For the Monte Carlo model, $\kappa = 1$, $\beta = 0.3$ is used.
2. The **neural-network ANE** is supported by finite-sample Monte Carlo evidence (Design 2) but falls outside the formal asymptotic theory. The gap arises from the interaction between the Lipschitz factor $1/\tau$ in the log-likelihood (Lemma 4.5) and the sieve error ε_n : condition (R3) requires $\varepsilon_n = o(\tau_n/\sqrt{n})$, which is stringent when $\tau_n \rightarrow 0$.
3. Relaxing condition (R3) via **Neyman orthogonality** (Chernozhukov et al., 2018; Farrell et al., 2021)—constructing a score that is first-order insensitive to ε_n —is a promising direction. If the score satisfies a Neyman orthogonality condition, the rate requirement weakens from $\sqrt{n}\varepsilon_n/\tau_n \rightarrow 0$ to $\sqrt{n}\varepsilon_n^2/\tau_n \rightarrow 0$, making feasibility possible. We leave this extension to future work.

Remark 5.6 (Computational motivation for neural-network sieves). *Despite the negative formal result, the neural-network sieve remains practically important. Grid-based value iteration costs $O(M^{d_s})$ evaluations per Bellman step, where M is the number of grid points per dimension. For $d_s = 5$ and $M = 50$, this is $50^5 \approx 3 \times 10^8$ evaluations—prohibitive for an inner loop embedded in optimisation over θ . A neural network with $K = 128$ units, $L = 4$ layers, and $n_{\text{iter}} = 1000$ fitted value-iteration steps requires $O(d_s K L n_{\text{iter}}) \approx 2.6 \times 10^6$ operations, two orders of magnitude faster. Bridging the gap between this computational advantage and the formal theory via Neyman orthogonality (point 3 above) is an important open problem.*

6 Asymptotic Theory

This section establishes the large-sample properties of the ANE estimator defined in (??). We show that, under the regularity, MZI, identification, and rate conditions stated in Sections 2–5, ANE is consistent and asymptotically normal with the same limiting distribution as the infeasible hard-max maximum likelihood estimator. In particular, the vanishing temperature does not affect the asymptotic variance: annealing is neutral in the sense of Proposition 4.7.

We treat both the cross-sectional i.i.d. case and the geometrically ergodic Markov case covered by Assumption 4. The Markov case requires replacing the usual variance by a long-run variance but is otherwise analogous.

6.1 Additional Assumptions

Assumption 8 (Score and Hessian regularity).

- (i) *The limiting score variance $\Omega := \lim_{\tau \downarrow 0} \text{Var}_{P_0}(\psi_\tau(Z; \theta_0))$ (Definition 3.7) exists and is positive definite. Under Assumption 4(b), replace Ω with the long-run variance $\Omega_{LR} := \lim_{\tau \downarrow 0} \sum_{k=-\infty}^{\infty} \text{Cov}(\psi_\tau(z_0; \theta_0), \psi_\tau(z_k; \theta_0))$.*
- (ii) *The limiting Hessian $H_0 := \lim_{\tau \downarrow 0} \nabla_{\theta\theta}^2 M(\theta_0, \tau)$ (Definition 3.7) exists and is negative definite.*
- (iii) **Margin condition:** *The margin condition (22) holds with exponent $\kappa \geq 2$.*
- (iv) **Lyapunov condition:** *There exists $\delta > 0$ such that $\sup_{\tau \in (0,1]} \mathbb{E}[\|\psi_\tau(Z; \theta_0)\|^{2+\delta}] < \infty$.*

Remark 6.1 (Necessity of $\kappa \geq 2$). *Assumption 8(iii) is not merely technical. Under the margin condition with exponent κ , the second moment of the annealed score satisfies $\mathbb{E}[\|\psi_\tau\|^2] = O(\tau^{\kappa-2})$ (Appendix C.5). For $\kappa < 2$, this diverges as $\tau \rightarrow 0$, and the standard $\sqrt{n}$ -normalisation fails to produce a non-degenerate limit. The restriction $\kappa \geq 2$ ensures that the score variance converges to a finite limit.*

The Lyapunov condition (iv) is verified in Appendix B.3 under $\kappa \geq 2 + \delta$. For $\kappa \in [2, 2 + \delta)$, condition (iv) must be imposed directly.

6.2 Consistency

Theorem 6.2 (Consistency). *Under Assumptions 5–7 and 8, the ANE estimator satisfies $\hat{\theta}_n \xrightarrow{P} \theta_0$.*

Proof. We verify the conditions of Newey and McFadden (1994, Theorem 2.1) for the criterion $\hat{Q}_n(\theta) := \hat{L}_n(\theta, \tau_n; \hat{V}_n)$ and the limiting criterion $Q_0(\theta) := \lim_{\tau \rightarrow 0} M(\theta, \tau)$.

- (a) **Identification:** θ_0 is the unique maximiser of Q_0 by Proposition 4.7.
- (b) **Compactness:** Θ is compact.

(c) **Uniform convergence:** Decompose:

$$\begin{aligned} \sup_{\theta} |\hat{Q}_n(\theta) - Q_0(\theta)| &\leq \underbrace{\sup_{\theta} |\hat{L}_n(\theta, \tau_n; \hat{V}_n) - \hat{L}_n(\theta, \tau_n; V_{\theta, \tau_n})|}_{R_1} + \underbrace{\sup_{\theta} |\hat{L}_n(\theta, \tau_n; V_{\theta, \tau_n}) - M(\theta, \tau_n)|}_{R_2} \\ &\quad + \underbrace{\sup_{\theta} |M(\theta, \tau_n) - Q_0(\theta)|}_{R_3}. \end{aligned}$$

$R_1 \rightarrow 0$: By Lemma 4.5, $R_1 \leq (2\beta/\tau_n)\varepsilon_n \rightarrow 0$ under (R3).

$R_2 \xrightarrow{p} 0$: For each fixed n , the class $\mathcal{F}_n := \{\ell(\cdot; \theta, \tau_n; V_{\theta, \tau_n}) : \theta \in \Theta\}$ is a Lipschitz image of the compact set Θ (the map $\theta \mapsto \ell(z; \theta, \tau_n; V_{\theta, \tau_n})$ is Lipschitz in θ for fixed $\tau_n > 0$, with Lipschitz constant $L(\tau_n) := C/\tau_n$). The covering number $N(\varepsilon, \mathcal{F}_n, L^1(P_n))$ is bounded by $N(\varepsilon\tau_n/C, \Theta, \|\cdot\|)$, which depends on n only through τ_n . Since Θ is compact, its covering numbers are finite, and by the Glivenko–Cantelli theorem for fixed $\mathcal{F}_n$:

$$R_2 = \sup_{\theta} \left| \frac{1}{n} \sum_{i=1}^n \ell(z_i; \theta, \tau_n; V_{\theta, \tau_n}) - \mathbb{E}[\ell(Z; \theta, \tau_n; V_{\theta, \tau_n})] \right| \xrightarrow{p} 0.$$

Under Assumption 4(b), the ergodic theorem replaces the LLN, and geometric ergodicity ensures uniform convergence by Andrews (1992, Theorem 3.1) applied to the class $\mathcal{F}_n$ for each n .

The triangular-array issue (the class changes with n) is handled by noting that R_2 decomposes as: for each n , R_2 is the uniform deviation of the empirical process over a fixed (for that n) class, which converges to zero in probability; we do not require uniform convergence over all n simultaneously.

$R_3 \rightarrow 0$: This follows from Proposition 4.7 and the uniform bound $|M(\theta, \tau)| \leq \log |\mathcal{A}|$ for all (θ, τ) , combined with the pointwise convergence $M(\theta, \tau) \rightarrow Q_0(\theta)$ and a finite covering of Θ .

All conditions of Newey and McFadden (1994, Theorem 2.1) hold. $\square$

6.3 Asymptotic Normality

Define the annealed score process at temperature τ_n ,

$$\psi_{\tau_n}(\theta; Z_t) = \nabla_{\theta} \ell_{\tau_n}(\theta; Z_t), \quad \Psi_n(\theta, \tau_n) = \frac{1}{n} \sum_{t=1}^n \psi_{\tau_n}(\theta; Z_t),$$

and let

$$\Sigma_0 := \lim_{n \rightarrow \infty} \text{Var}_{P_0} \left(n^{-1/2} \sum_{t=1}^n \psi_{\tau_n}(\theta_0; Z_t) \right)$$

denote the asymptotic variance of the score process. Under Assumption 4(a) (i.i.d.), $\Sigma_0 = \Omega$; under Assumption 4(b) (Markov), $\Sigma_0 = \Omega_{LR}$.

Theorem 6.3 (Asymptotic normality). *Under Assumptions 5–8,*

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, H_0^{-1} \Sigma_0 H_0^{-1}), \quad (31)$$

where H_0 and Ω are the limits defined in (14)–(15), and $\Sigma_0 = \Omega_{LR}$ under Assumption 4(b).

Proof. The proof follows six steps.

Step 1: Score decomposition. By a mean-value expansion:

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = - \left[\nabla_{\theta\theta}^2 \hat{L}_n(\tilde{\theta}_n, \tau_n; \hat{V}_n) \right]^{-1} \sqrt{n} \nabla_{\theta} \hat{L}_n(\theta_0, \tau_n; \hat{V}_n), \quad (32)$$

where $\tilde{\theta}_n$ lies between $\hat{\theta}_n$ and θ_0 .

Step 2: Decompose the score at θ_0 .

$$\begin{aligned} \sqrt{n} \nabla_{\theta} \hat{L}_n(\theta_0, \tau_n; \hat{V}_n) &= \underbrace{\frac{1}{\sqrt{n}} \sum_{i=1}^n [\psi_{\tau_n}(z_i; \theta_0) - \mathbb{E}[\psi_{\tau_n}(Z; \theta_0)]]}_{=: S_n} \\ &+ \underbrace{\sqrt{n} [\bar{\psi}_{\tau_n}(\hat{V}_n) - \bar{\psi}_{\tau_n}(V_{\theta_0, \tau_n})]}_{=: A_n} + \underbrace{\sqrt{n} \mathbb{E}[\psi_{\tau_n}(Z; \theta_0)]}_{=: B_n}. \end{aligned} \quad (33)$$

Step 3: $A_n = o_p(1)$ (approximation bias). By the Lipschitz bound for the score in V (which has constant $C\beta/\tau_n$, derived analogously to Lemma 4.5): $\|A_n\| \leq \sqrt{n} \cdot (C\beta/\tau_n) \cdot \varepsilon_n \rightarrow 0$ under (R3).

Step 4: $B_n = o(1)$ (smoothing bias). $\mathbb{E}[\psi_{\tau_n}(Z; \theta_0)] = \nabla_{\theta} M(\theta_0, \tau_n)$. By Proposition 4.4 (applied to the derivative), $\|\nabla_{\theta} M(\theta_0, \tau_n)\| \leq C' \tau_n^{\kappa} [\log(1/\tau_n)]^{\kappa}$. Under (R4), $\sqrt{n} \cdot \tau_n^{\kappa} \rightarrow 0$, so $\|B_n\| \rightarrow 0$.

Step 5: $S_n \Rightarrow N(0, \Sigma_0)$ (CLT for triangular array).

The scores $\{\psi_{\tau_n}(z_i; \theta_0)\}_{i=1}^n$ form a triangular array (since τ_n changes with n).

Variance convergence. Let $\Omega_n := \text{Var}(\psi_{\tau_n}(Z; \theta_0))$. By Assumption 8(i) and Lemma 4.6, $\Omega_n \rightarrow \Omega$ as $n \rightarrow \infty$ (since $\tau_n \rightarrow 0$).

Lyapunov condition. By Assumption 8(iv), $\sup_n \mathbb{E}[\|\psi_{\tau_n}\|^{2+\delta}] \leq C < \infty$. The Lyapunov ratio is $n^{-1} \mathbb{E}[\|\psi_{\tau_n}\|^{2+\delta}] / (\text{Var}(\psi_{\tau_n}))^{(2+\delta)/2} \leq C / (n \lambda_{\min}(\Omega_n))^{(2+\delta)/2}$. Since $\Omega_n \rightarrow \Omega \succ 0$, the denominator is bounded below and the ratio goes to zero.

By the Lyapunov CLT for triangular arrays, $S_n \Rightarrow N(0, \Omega)$.

Under Assumption 4(b), the CLT for geometrically ergodic chains (Jones and Hobert, 2001) applies, yielding $S_n \Rightarrow N(0, \Omega_{LR})$; see also Meyn and Tweedie (2009), Chapter 17.

Step 6: Hessian convergence. We show $\hat{H}_n := \nabla_{\theta\theta}^2 \hat{L}_n(\tilde{\theta}_n, \tau_n; \hat{V}_n) \xrightarrow{p} H_0$.

Decompose:

$$\begin{aligned} \hat{H}_n - H_0 &= \underbrace{[\nabla^2 \hat{L}_n(\tilde{\theta}_n, \tau_n; \hat{V}_n) - \nabla^2 \hat{L}_n(\tilde{\theta}_n, \tau_n; V_{\theta_0, \tau_n})]}_{H_1} \\ &+ \underbrace{[\nabla^2 \hat{L}_n(\tilde{\theta}_n, \tau_n; V_{\theta_0, \tau_n}) - \nabla^2 \hat{L}_n(\theta_0, \tau_n; V_{\theta_0, \tau_n})]}_{H_2} \\ &+ \underbrace{[\nabla^2 \hat{L}_n(\theta_0, \tau_n; V_{\theta_0, \tau_n}) - \mathbb{E}[\nabla^2 \ell(Z; \theta_0, \tau_n; V_{\theta_0, \tau_n})]]}_{H_3} \\ &+ \underbrace{[\mathbb{E}[\nabla^2 \ell(Z; \theta_0, \tau_n; V_{\theta_0, \tau_n})] - H_0]}_{H_4}. \end{aligned}$$

$H_1 = o_p(1)$: By a second-order Lipschitz bound, $\|H_1\| \leq C\varepsilon_n/\tau_n^2 = Cn^{-\alpha+2\beta}$. Under (R3) and $\beta < 1/2$: $\alpha > \beta + 1/2 > 2\beta$ (since $\beta < 1/2$ implies $\beta + 1/2 > 2\beta$), so $-\alpha + 2\beta < 0$ and $H_1 \rightarrow 0$.

$H_2 = o_p(1)$: By the smoothness of the Hessian in θ for fixed $\tau_n > 0$ and $\tilde{\theta}_n \rightarrow \theta_0$.

$H_3 = o_p(1)$: LLN for a fixed (for each n) function class, analogous to the R_2 argument in Theorem 6.2. The effective variance of the Hessian is $O(\tau_n^{\kappa-4})$; for $\kappa \geq 2$ and the Bernstein

inequality with $\beta < 1/2$, the exponent in the concentration bound is $n \cdot \tau_n^{4-\kappa} = n^{1-(4-\kappa)\beta}$. For $\kappa \geq 2$ and $\beta < 1/2$: $1 - (4 - \kappa)\beta \geq 1 - 2\beta > 0$, so the concentration probability goes to zero.

$H_4 = o(1)$: By Theorem 3.8(i), $\nabla^2 M(\theta_0, \tau_n) \rightarrow H_0$.

Combining Steps 1–6: $\sqrt{n}(\hat{\theta}_n - \theta_0) = -H_0^{-1}S_n + o_p(1) \Rightarrow N(0, H_0^{-1}\Sigma_0 H_0^{-1})$. $\square$

6.4 Discussion

Theorem 6.3 shows that, under the MZI property, vanishing temperature, and appropriate rate conditions, ANE has standard $\sqrt{n}$ -Gaussian asymptotics. The cube-root phenomena associated with non-smooth M-estimation (Kim and Pollard, 1990) do not arise here, because the non-smoothness of the Bellman operator is confined to a measure-zero manifold and the population criterion is smooth at θ_0 .

Efficiency

The asymptotic variance $H_0^{-1}\Sigma_0 H_0^{-1}$ equals that of the *limit of the soft-max MLE* as $\tau \rightarrow 0$. There is no well-defined “hard-max MLE” in the usual sense: the hard-max likelihood is degenerate (0 or $-\infty$), so the comparison must be understood as the limit of the annealed estimator’s variance.

We do *not* prove semiparametric efficiency. ANE is at least as efficient as any estimator using the same limiting score. Establishing whether $H_0^{-1}\Sigma_0 H_0^{-1}$ coincides with the semiparametric efficiency bound requires characterising the tangent space of the model with kinks and verifying that the influence function lies in it—a technically demanding question that we leave for future work (see Remark 3.12).

6.5 Variance Estimation

A consistent estimator of the asymptotic variance is

$$\widehat{\text{Var}}(\hat{\theta}_n) = \hat{H}_n^{-1} \hat{\Sigma}_n \hat{H}_n^{-1}, \quad (34)$$

where $\hat{H}_n = \nabla^2 \hat{L}_n(\hat{\theta}_n, \tau_n; \hat{V}_n)$ and $\hat{\Sigma}_n = n^{-1} \sum_{i=1}^n \psi_{\tau_n}(z_i; \hat{\theta}_n) \psi_{\tau_n}(z_i; \hat{\theta}_n)^\top$. Under Assumption 4(b), $\hat{\Sigma}_n$ should be replaced by a HAC estimator (Newey and West, 1987). In the i.i.d. case, the plug-in formula above suffices.

7 Computational Algorithm

This section describes practical implementations of the ANE for finite samples. We present a unified algorithmic framework that accommodates both grid-based approximation (suitable for $d_s \leq 3$) and neural-network sieves (required for $d_s \geq 4$). Section 7.1 outlines the overall structure. Section 7.2 details value function approximation via grid-based and neural-network fitted value iteration. Section 7.3 describes outer-loop optimisation with annealing. The algorithms presented here have been implemented and tested in the Monte Carlo experiments of Section 8.

7.1 Algorithmic overview

For a given sample $\{(S_t, A_t)\}_{t=1}^n$ and a decreasing temperature path $\{\tau_k\}_{k=1}^K$ with $\tau_K = \tau_n$, the ANE algorithm consists of nested optimisation:

Outer loop (over θ):

For each temperature τ_k , maximise the annealed log-likelihood $\hat{L}_n(\theta, \tau_k; \hat{V}_{\theta, \tau_k})$ over $\theta \in \Theta$ using gradient-based optimisation.

Inner loop (fixed-point iteration):

For each (θ, τ_k) evaluated in the outer loop, approximate the fixed point V_{θ, τ_k} of the annealed Bellman operator T_θ^τ using fitted value iteration in a suitable approximation space.

The temperature path provides a continuation strategy: we begin with a high temperature τ_1 (where the problem is smooth and nearly convex), solve for $\hat{\theta}_1$, then decrease temperature gradually, warm-starting each stage from the previous solution. The final temperature $\tau_K = \tau_n$ is determined by the sample size n and the rate conditions (R1)–(R4) from Section 5.3.

Comparison with existing methods. The ANE algorithm differs from standard dynamic discrete choice estimators in two key respects. First, unlike NFXP (Rust, 1987) or MPEC (Su and Judd, 2012), which solve the hard-max Bellman equation ($\tau = 0$), ANE solves a smoothed version ($\tau = \tau_n > 0$) that is differentiable everywhere, eliminating kinks in the likelihood and enabling efficient gradient-based optimisation. Second, unlike CCP-based methods (Hotz and Miller, 1993) that require invertibility and terminal-period assumptions, ANE approximates the value function directly using either grids or neural networks, with the approximation error entering the asymptotics transparently through the rate conditions. The relationship to grid-based NFXP is particularly close: ANE can be viewed as “smoothed NFXP” where the hard-max in the Bellman operator is replaced by log-sum-exp.

7.2 Value function approximation

The inner loop approximates the fixed point $V_{\theta, \tau}$ satisfying $(T_\theta^\tau V_{\theta, \tau})(s) = V_{\theta, \tau}(s)$ where T_θ^τ is the annealed Bellman operator defined in (16).

7.2.1 Grid-based fitted value iteration

Setup. Let $\mathcal{S}_M = \{s^{(1)}, \dots, s^{(M)}\}$ be a grid of M points. For $d_s \leq 3$, tensor-product grids with $M = \prod_{j=1}^{d_s} M_j$ are feasible. Represent V as a vector $\mathbf{V} \in \mathbb{R}^M$ and interpolate between grid points using piecewise linear or cubic splines.

Q-values. For each action $a \in \mathcal{A}$, construct the $M \times M$ transition matrix $P_\theta^{(a)}$ and compute

$$Q_{\theta, a}[i] = u(s^{(i)}, a; \theta) + \beta \sum_{j=1}^M P_\theta^{(a)}[i, j] \cdot V[j], \quad i = 1, \dots, M. \quad (35)$$

This is a matrix-vector product $Q_{\theta, a} = u_{\theta, a} + \beta P_\theta^{(a)} \mathbf{V}$.

Bellman update. Apply the annealed Bellman operator using the log-sum-exp trick for numerical stability:

$$(T_\theta^\tau V)[i] = \max_a Q_{\theta, a}[i] + \tau \log \sum_{a \in \mathcal{A}} \exp\left\{ \frac{Q_{\theta, a}[i] - \max_{a'} Q_{\theta, a'}[i]}{\tau} \right\}. \quad (36)$$

Iteration. Initialise $V^{(0)} = \mathbf{0}$ (or warm start) and iterate $V^{(\ell+1)} = T_\theta^\tau V^{(\ell)}$ until $\|V^{(\ell+1)} - V^{(\ell)}\|_\infty < \delta_n$. By Lemma 4.1, convergence is geometric: $\|V^{(\ell)} - V_{\theta, \tau}\|_\infty \leq \beta^\ell \|V^{(0)} - V_{\theta, \tau}\|_\infty$. In practice, $\ell \leq 100$ iterations suffice to reach $\delta_n = 10^{-6}$ from a warm start.

Algorithm 1 Grid-based fitted value iteration at (θ, τ)

Require: Grid $\mathcal{S}_M$, transition matrices $\{P_\theta^{(a)}\}_{a \in \mathcal{A}}$, parameters (θ, τ, β) , tolerance δ_n

```
1: Initialise  $V^{(0)} \in \mathbb{R}^M$  (zero or warm start)
2: for  $\ell = 0, 1, \dots, \ell_{\max}$  do
3:   for  $a \in \mathcal{A}$  do
4:      $Q_{\theta,a} \leftarrow u_{\theta,a} + \beta P_\theta^{(a)} V^{(\ell)}$ 
5:   end for
6:    $V^{(\ell+1)}[i] \leftarrow \tau \log \sum_a \exp\{Q_{\theta,a}[i]/\tau\}$  via (36) for  $i = 1, \dots, M$ 
7:   if  $\|V^{(\ell+1)} - V^{(\ell)}\|_\infty < \delta_n$  then
8:     break
9:   end if
10: end for
11: return  $\hat{V}_{\theta,\tau} = V^{(\ell+1)}$ 
```

Remark 7.1 (Grid computational cost). *Each FVI iteration requires $O(M^2|\mathcal{A}|)$ operations for the matrix-vector products. For tensor-product grids, $M = M_0^{d_s}$ grows exponentially in dimension: $d_s = 1$ with $M = 200$ takes $\approx 0.01s$ per iteration; $d_s = 3$ with $M = 100^3 = 10^6$ takes $\approx 10s$; $d_s = 5$ with $M = 50^5 \approx 3 \times 10^8$ is prohibitive.*

7.2.2 Neural-network fitted value iteration

For $d_s \geq 4$, we parameterise V as a feedforward ReLU network V_ϕ with architecture (26), depth $L_n = O(\log n)$, and width $K_n = O(n^{d_s/(2r+d_s)})$ as in Section 5.1.

State sampling. Since we cannot evaluate the Bellman operator at all $s \in \mathcal{S}$, we approximate using M state nodes. We draw from an approximation to the stationary distribution by simulating the model under preliminary parameter estimates for $T = 10,000$ periods after a burn-in, and subsample $M = 2,000$ states uniformly.

Bellman targets. For each state node $s^{(m)}$, compute

$$y_\ell^{(m)} = \tau \log \sum_{a \in \mathcal{A}} \exp \left\{ \frac{u(s^{(m)}, a; \theta) + \beta \frac{1}{N_{\text{sim}}} \sum_{j=1}^{N_{\text{sim}}} V_{\phi^{(\ell)}}(g_\theta(s^{(m)}, a, \varepsilon_j))}{\tau} \right\}, \quad (37)$$

where $\{\varepsilon_j\}_{j=1}^{N_{\text{sim}}}$ are i.i.d. draws from the shock distribution ($N_{\text{sim}} = 100$).

Regression step. Update network parameters by minimising the squared Bellman error:

$$\phi^{(\ell+1)} \in \underset{\phi}{\operatorname{argmin}} \frac{1}{M} \sum_{m=1}^M (V_\phi(s^{(m)}) - y_\ell^{(m)})^2 + \lambda_n R(\phi), \quad (38)$$

where $R(\phi)$ is an ℓ_2 weight-decay penalty. Minimisation uses Adam (Kingma and Ba, 2015) for $E = 50$ epochs. Iterate until $\max_m |V_{\phi^{(\ell+1)}}(s^{(m)}) - V_{\phi^{(\ell)}}(s^{(m)})| < \delta_n$.

Algorithm 2 Neural-network fitted value iteration at (θ, τ)

Require: State nodes $\{s^{(m)}\}_{m=1}^M$, architecture (L_n, K_n) , parameters (θ, τ, β) , tolerance δ_n

```
1: Initialise  $\phi^{(0)}$  (Xavier or warm start)
2: for  $\ell = 0, 1, \dots, \ell_{\max}$  do
3:   for  $m = 1, \dots, M$  and  $a \in \mathcal{A}$  do
4:     Compute Q-value via Monte Carlo integration (37)
5:   end for
6:   Fit  $\phi^{(\ell+1)}$  via (38) using Adam ( $E$  epochs)
7:   if  $\max_m |V_{\phi^{(\ell+1)}}(s^{(m)}) - V_{\phi^{(\ell)}}(s^{(m)})| < \delta_n$  then
8:     break
9:   end if
10: end for
11: return  $\hat{V}_{\theta, \tau} = V_{\phi^{(\ell+1)}}$ 
```

Remark 7.2 (Neural-network computational cost). *Each FVI iteration requires $O(M|\mathcal{A}|N_{\text{sim}} \cdot L_n K_n)$ operations for forward passes plus $O(E \cdot M \cdot L_n K_n)$ for SGD fitting. This scales linearly in d_s (through the input dimension), not exponentially as with grids. For $d_s = 5$, $M = 2000$, $N_{\text{sim}} = 100$, $L_n = 4$, $K_n = 128$, each iteration takes $\approx 5s$ on a GPU—two to three orders of magnitude faster than grid-based methods at the same dimension (Remark 5.6).*

7.3 Outer-loop optimisation

7.3.1 Annealed log-likelihood and score

Given the approximate value function $\hat{V}_{\theta, \tau}$, compute conditional choice probabilities

$$p_{\theta, \tau}(a | s) = \frac{\exp\{Q_{\theta}(s, a; \hat{V}_{\theta, \tau})/\tau\}}{\sum_{a'} \exp\{Q_{\theta}(s, a'; \hat{V}_{\theta, \tau})/\tau\}}, \quad (39)$$

and the sample log-likelihood $\hat{L}_n(\theta, \tau) = n^{-1} \sum_{t=1}^n \log p_{\theta, \tau}(A_t | S_t)$.

The score contribution from observation t is

$$\psi_n(Z_t; \theta, \tau) = \frac{1}{\tau} \left[\nabla_{\theta} Q_{\theta}(S_t, A_t; \hat{V}_{\theta, \tau}) - \sum_{a'} p_{\theta, \tau}(a' | S_t) \nabla_{\theta} Q_{\theta}(S_t, a'; \hat{V}_{\theta, \tau}) \right]. \quad (40)$$

Practical simplification. We treat $\hat{V}_{\theta, \tau}$ as *fixed* when computing $\nabla_{\theta} Q_{\theta}$, setting $\nabla_{\theta} \hat{V}_{\theta, \tau} = 0$. This is justified asymptotically: the indirect effect through $\nabla_{\theta} \hat{V}_{\theta, \tau}$ contributes an additional $O(\varepsilon_n)$ term that is dominated by the direct score, and the sieve error is already controlled by the rate conditions (R1)–(R3). The simplification reduces computational cost significantly, avoiding the need to differentiate through the entire FVI procedure.

7.3.2 Optimisation and annealing schedule

We use L-BFGS with line search to maximise $\hat{L}_n(\theta, \tau_k)$ over $\theta \in \Theta$ at each temperature stage. The temperature path $\{\tau_1, \dots, \tau_K\}$ is geometric:

$$\tau_k = \tau_1 \cdot \gamma^{k-1}, \quad k = 1, \dots, K, \quad \gamma = (\tau_K / \tau_1)^{1/(K-1)}. \quad (41)$$

Practical guidance. $\tau_1 = 1$ (fully smooth softmax); decay factor $\gamma \in [0.5, 0.8]$; final temperature $\tau_K = n^{-\beta}$ with $\beta \in (1/(2\kappa), 1/2)$ from Assumption 7; number of stages $K = \lceil \log(\tau_1/\tau_K)/\log(1/\gamma) \rceil$, typically $K = 4-8$.

Remark 7.3 (Data-dependent annealing). *The implementation ties τ_K to n via $\tau_K = c \cdot n^{-\beta}$. In the Monte Carlo, $c = 1$ and $\beta = 0.25$, giving $\tau_K \in \{0.18, 0.15, 0.12, 0.10\}$ for $n \in \{1000, 2000, 5000, 10000\}$. For the neural-network design, $\beta = 0.30$ gives slightly faster annealing: $\tau_K \in \{0.13, 0.10, 0.08, 0.06\}$.*

7.3.3 Full algorithm

Algorithm 3 combines the inner and outer loops.

Algorithm 3 ANE with grid or neural-network sieve

Require: Data $\{(S_t, A_t)\}_{t=1}^n$; schedule $\tau_1 > \dots > \tau_K > 0$; approximation method (grid or NN)

- 1: Initialise $\theta^{(0)}$
 - 2: **for** $k = 1, \dots, K$ **do**
 - 3: Set $\tau \leftarrow \tau_k$
 - 4: Solve for $\hat{V}_{\theta^{(k-1)}, \tau}$ via Algorithm 1 or 2
 - 5: Maximise $\hat{L}_n(\theta, \tau; \hat{V}_{\theta, \tau})$ over θ using L-BFGS, warm-started at $\theta^{(k-1)}$
 - 6: Set $\theta^{(k)} \leftarrow \hat{\theta}$
 - 7: **end for**
 - 8: **Return** $\hat{\theta}_n := \theta^{(K)}$; compute $\widehat{\text{Var}}$ via (34)
-

7.4 Practical considerations

Remark 7.4 (Parameter constraints). *When structural parameters must satisfy economic restrictions (e.g., $\alpha > 0$, $\gamma \in [1, 5]$), we reparameterise using unconstrained auxiliary variables $\phi \in \mathbb{R}^{d_\theta}$ and a smooth bijection $\theta = g(\phi)$. For positivity constraints, use $\theta_j = \theta_j^{\min} + \exp(\phi_j)$; for bounded intervals, use the sigmoid $\theta_j = \theta_j^{\min} + (\theta_j^{\max} - \theta_j^{\min})\sigma(\phi_j)$. Reparameterisation does not affect the asymptotic distribution: by the delta method, standard errors are transformed via the Jacobian $J_g(\phi_0)$.*

Remark 7.5 (Diagnostics). *Applied implementations benefit from three diagnostic checks: (i) Bellman residuals—at each τ_k , compute $R_{\theta, \tau_k} := \max_m |\hat{V}_{\theta, \tau_k}(s^{(m)}) - (T_\theta^\tau \hat{V}_{\theta, \tau_k})(s^{(m)})|$ on a held-out set; values below 10^{-4} indicate accurate FVI; (ii) stability across architectures—re-estimate with different network depths ($L_n \in \{3, 4, 5\}$) and widths ($K_n \in \{64, 128, 256\}$); stable estimates suggest well-calibrated approximation; (iii) grid-based benchmarking—for $d_s \leq 2$, compare ANE-NN to ANE-grid (Algorithm 1) to verify that sieve error is negligible.*

8 Monte Carlo Evidence

This section reports Monte Carlo experiments that assess the finite-sample performance of the ANE relative to standard alternatives. We consider two designs based on a dynamic labour-supply model. In both designs, the data-generating process (DGP) is the hard-max model at the true parameter $\theta_0 = (\alpha_0, \gamma_0) = (1.8, 2.0)$, and we simulate a Markov chain from the optimal hard-max policy. Design 1 uses grid-based value iteration and treats ANE as the benchmark estimator in a setting with no sieve approximation error. Design 2 replaces the grid by a neural-network sieve and compares ANE with fixed-softmax NPL and CCP-type estimators.

Throughout, we report Monte Carlo bias and root mean squared error (RMSE) for the structural parameters (α, γ) across sample sizes $n \in \{1000, 2000, 5000\}$ with $R = 200$ replications.

8.1 Model and data-generating process

In each period t , the agent observes a continuous state $s_t \in \mathcal{S} = [s_{\min}, s_{\max}]$ representing liquid assets and chooses $a_t \in \{0, 1\}$, where $a_t = 1$ denotes work and $a_t = 0$ denotes leisure. Per-period utility is

$$u(s, a; \theta) = \begin{cases} \frac{(s + wa)^{1-\gamma} - 1}{1 - \gamma} - \alpha a, & \text{if } \gamma \neq 1, \\ \log(s + wa) - \alpha a, & \text{if } \gamma = 1, \end{cases}$$

where $w > 0$ is the wage, $\alpha > 0$ is the disutility of work, and $\gamma > 0$ is the coefficient of relative risk aversion. The state evolves according to

$$s_{t+1} = \rho s_t + wa_t + \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, \sigma^2),$$

with reflecting boundaries at $[s_{\min}, s_{\max}]$ and discount factor $\beta \in (0, 1)$.

We calibrate the primitives to values standard in the labour literature:

$$\beta = 0.95, \quad \rho = 0.85, \quad w = 1.0, \quad \sigma = 0.4, \quad \mathcal{S} = [0.1, 10.0],$$

and set the true structural parameters to $\theta_0 = (\alpha_0, \gamma_0) = (1.8, 2.0)$.

The optimal policy at θ_0 is a one-sided threshold rule: there exists a unique cutoff $s^*(\theta_0) \in (0.1, 10)$ such that $a_{\theta_0}^*(s) = 1$ if $s < s^*(\theta_0)$ and $a_{\theta_0}^*(s) = 0$ if $s > s^*(\theta_0)$.

Simulation procedure. For each replication, we simulate a single Markov path of length $T_{\text{burn}} + n$ with $T_{\text{burn}} = 300$ and sample size $n \in \{1000, 2000, 5000\}$. The initial state s_1 is drawn from $\text{Unif}[0.1, 10.0]$, and the agent follows the hard-max policy $a_t = a_{\theta_0}^*(s_t)$ at every date. We discard the first T_{burn} observations and retain $(S_t, A_t)_{t=1}^n$ for estimation. To implement $a_{\theta_0}^*$ on the continuous state space, we solve the Bellman equation on a fine grid of $M_{\text{DGP}} = 200$ points and use nearest-neighbour interpolation.

8.2 Estimators

All estimators treat (S_t, A_t) as generated by a dynamic discrete choice model with the primitives above and structural parameter $\theta = (\alpha, \gamma)$. We always initialise at $(\alpha, \gamma) = (1.0, 1.5)$ for comparability across methods.

- (i) **ANE-grid (Design 1).** We discretise $\mathcal{S}$ on a grid of $M = 150$ points and implement ANE as in Section 4. The annealed Bellman operator uses the log-sum-exp mapping, and value functions are solved by value iteration with sup-norm tolerance 10^{-6} . The annealing schedule is $\tau_n = n^{-\beta}$ with $\beta = 0.25$, lying in the feasible interval from Assumption 7. The temperature path consists of $K = 8$ geometrically spaced values between $\tau_1 = 1.0$ and τ_n . At each temperature, we maximise the annealed log-likelihood over θ using Nelder–Mead, and use the final estimate at τ_n as the ANE-grid estimator. This design has no sieve approximation error and provides a clean benchmark for the large-sample theory.
- (ii) **ANE-NN (Design 2).** We replace the grid by a feedforward value network $V(s; \phi)$ with three hidden layers and 128 Tanh units per layer. For each (θ, τ) , we update ϕ by minimising the mean squared Bellman residual using Adam (batch size 256). The temperature schedule is geometric between $\tau_1 = 1.0$ and $\tau_n = n^{-\beta}$ with $\beta = 0.30$ and $K = 8$ stages.

Table 3: Monte Carlo results: Design 1 (grid-based ANE)

Estimator	Parameter	$n = 1000$		$n = 2000$		$n = 5000$	
		Bias	RMSE	Bias	RMSE	Bias	RMSE
ANE-grid	α	0.232	0.233	0.210	0.211	0.183	0.183
	γ	0.096	0.103	0.055	0.060	0.018	0.022

Notes: True parameters are $(\alpha_0, \gamma_0) = (1.8, 2.0)$. Results based on $R = 200$ replications per sample size. ANE-grid uses $\tau_n = n^{-0.25}$ with $K = 8$ annealing stages and grid-based value iteration on $M = 150$ points.

- (iii) **NPL-NN (fixed softmax).** Nested Pseudo-Likelihood iterations as in Aguirregabiria and Mira (2002), implemented with a fixed logit temperature $\tau \in \{0.5, 0.3, 0.1\}$ throughout estimation. Because τ is held constant as n grows, these estimators target pseudo-true parameters $\tilde{\theta}(\tau) \neq \theta_0$ whenever the DGP is hard-max.
- (iv) **CCP-NN.** A two-step CCP estimator in the spirit of Hotz and Miller (1993): estimate $p(a = 1 \mid s)$ in the first stage by a feedforward network trained via logistic regression, then recover θ in the second stage from a structural logit index. CCP estimators can perform poorly when the true policy has sharp thresholds, because the nonparametric first stage smooths out the discontinuity.

8.3 Design 1: grid-based benchmark

In Design 1, the value function is solved exactly on the grid and ANE-grid is the only estimator. Table 3 reports Monte Carlo bias and RMSE for (α, γ) across sample sizes.

Bias in γ is small and shrinks as n increases, consistent with the theoretical result that ANE targets the hard-max likelihood as $\tau_n \rightarrow 0$. Bias in α is more persistent and moderately positive, which we interpret as a finite-sample consequence of using a conservative annealing rate. Despite this, ANE-grid is numerically stable and converges reliably across replications, providing a clean benchmark when the Bellman operator is solved exactly.

8.4 Design 2: neural-network approximation

Design 2 replaces grid-based value iteration by a neural-network sieve while keeping the same hard-max DGP. We compare ANE-NN to NPL-NN with fixed temperatures $\tau \in \{0.5, 0.3, 0.1\}$ and to CCP-NN. Table 4 reports bias and RMSE for (α, γ) at $n = 5000$; results for smaller sample sizes display similar patterns and are reported in the online appendix.

Four patterns emerge from Table 4.

- (1) **ANE-NN remains reasonably accurate under sieve approximation.** At $n = 5000$, ANE-NN yields $(\hat{\alpha}, \hat{\gamma}) = (1.752, 2.229)$, corresponding to biases of approximately -0.05 for α and 0.23 for γ . The bias in α is modest relative to the true value (2–3%), while the bias in γ is more pronounced ($\approx 11\%$), reflecting the sensitivity of the risk-aversion parameter to approximation error in the continuation values. The RMSE of ANE-NN decreases with n , consistent with the large-sample theory in Section 6.
- (2) **Fixed-softmax NPL-NN exhibits pseudo-true bias.** For $\tau = 0.5$, NPL-NN produces nearly unbiased estimates of α (bias 0.023) but substantially overestimates γ (bias 0.310).

Table 4: Monte Carlo results: Design 2 (neural-network value approximation), $n = 5000$

Estimator	Parameter	Mean	Bias	RMSE	95% Cov.
ANE-NN	α	1.7524	-0.0476	0.0627	—
	γ	2.2286	0.2286	0.2321	—
NPL-NN ($\tau = 0.5$)	α	1.8230	0.0230	0.0237	—
	γ	2.3099	0.3099	0.3099	—
NPL-NN ($\tau = 0.3$)	α	1.7136	-0.0864	0.0932	—
	γ	2.2352	0.2352	0.2355	—
NPL-NN ($\tau = 0.1$)	α	1.5007	-0.2993	0.3013	—
	γ	2.0075	0.0075	0.0416	—
CCP-NN	α	0.8438	-0.9562	0.9563	—
	γ	2.1861	0.1861	0.1885	—

Notes: True parameters are $(\alpha_0, \gamma_0) = (1.8, 2.0)$. Results based on $R = 200$ replications with $n = 5000$. ANE-NN uses $\tau_n = n^{-0.30}$ with $K = 8$ annealing stages and a value network with three hidden layers of width 128. NPL-NN uses the same architecture but keeps τ fixed. CCP-NN estimates CCPs by a neural network in the first stage.

As τ decreases to 0.1, the estimator becomes nearly unbiased in γ (bias 0.008) but severely underestimates α (bias -0.299). No single fixed temperature dominates: τ trades bias in level and curvature in a way that reflects the pseudo-true behaviour of the fixed-softmax model rather than the hard-max target.

- (3) **CCP-NN fails to recover the threshold parameter.** CCP-NN grossly understates the disutility of work, with $\hat{\alpha} \approx 0.84$ and bias -0.96 , while displaying a moderate upward bias in γ . Almost all of the RMSE in α is squared bias. The nonparametric first stage smooths the discrete jump in the true policy, and the second-stage inversion then attributes this smoothed transition to extreme log-odds, leading to severe distortion of the structural parameters.
- (4) **Comparison with Design 1.** Relative to grid-based ANE, ANE-NN displays larger biases, especially in γ , reflecting the presence of sieve approximation error (Remark 5.5). Nonetheless, the qualitative behaviour aligns with the theory: bias decreases with n , and the estimator targets the hard-max model rather than a smoothed pseudo-true parameter.

8.5 Discussion

The Monte Carlo evidence supports the main theoretical insights of the paper. In the grid-based Design 1, ANE-grid provides a numerically stable estimator that targets the hard-max likelihood and exhibits small, shrinking bias in the threshold-relevant parameter γ . In the neural design, ANE-NN retains these qualitative properties despite approximation error. In both designs, estimators that impose permanent smoothing (fixed-softmax NPL) or rely on nonparametric CCP inversion display economically meaningful pseudo-true bias when probability mass concentrates near the participation threshold.

From a practical standpoint, these results suggest that annealing is a useful device for preserving the geometry of hard-max decision rules while leveraging the computational benefits of smooth objectives. In low-dimensional settings, ANE-grid can serve as a benchmark estimator comparable to classical NFXP. In higher-dimensional or more complex models, ANE-NN offers a viable alternative to fixed-softmax policy-iteration and CCP methods.

Remark 8.1 (Coverage of the formal theory). *The Monte Carlo model has a margin exponent $\kappa \approx 1$ (the invariant density is roughly uniform near the threshold), which is below the $\kappa \geq 2$ required by Assumption 8(iii). Consequently, the formal $\sqrt{n}$ asymptotic normality result (Theorem 6.3) does not directly apply to the Monte Carlo DGP. Nonetheless, ANE performs well empirically, suggesting that the $\kappa \geq 2$ condition may be an artefact of the proof technique (specifically, the crude bound on the score variance near the threshold) rather than a genuine barrier. Extending the formal theory to $\kappa < 2$ —perhaps via a different normalisation or a local asymptotic analysis—is an important direction for future work.*

A Proofs of Main Results

This appendix collects proofs deferred from the main text. Section A.1 establishes the manifold structure of indifference sets. Section A.2 gives the full MZI proof. Section A.3 records a local-uniqueness lemma used throughout. Sections A.4–A.5 prove directional differentiability of the Bellman fixed point and of integrated population criteria. Section A.6 proves existence of the annealed Hessian and variance limits. Section A.7 verifies annealing neutrality.

A.1 Manifold structure of indifference sets

Proposition A.1 (Indifference sets are C^1 submanifolds, restated). *Under Assumptions 5 and 1, for each θ in a neighbourhood of θ_0 and each pair (a, a') with $a \neq a'$:*

- (i) *The pairwise indifference set $\mathcal{K}_{a,a'}(\theta) := \{s \in \mathcal{S} : \Delta_{a,a'}(s; \theta) = 0\}$ is either empty or a C^1 embedded submanifold of $\mathbb{R}^{d_s}$ of dimension $d_s - 1$.*
- (ii) *The full indifference set $\mathcal{K}(\theta) = \bigcup_{a \neq a'} \mathcal{K}_{a,a'}(\theta)$ is a finite union of such submanifolds.*

Proof. Fix θ and (a, a') with $\mathcal{K}_{a,a'}(\theta) \neq \emptyset$. Define $\phi : \mathcal{S} \rightarrow \mathbb{R}$ by $\phi(s) = \Delta_{a,a'}(s; \theta)$. By Assumption 1(i), ϕ is C^1 . By Assumption 1(ii), $\nabla_s \phi(s) \neq 0$ for every $s \in \phi^{-1}(0)$, so 0 is a regular value of ϕ . The Regular Value Theorem (Guillemin and Pollack, 1974, Theorem 1.3.2) implies $\phi^{-1}(0)$ is a C^1 embedded submanifold of codimension 1, hence dimension $d_s - 1$. Since $|\mathcal{A}| < \infty$, the union $\mathcal{K}(\theta)$ is finite. $\square$

Remark A.2 (Geometric dimension). *For $d_s = 1$, indifference sets are isolated points (the thresholds of Section 2.2). For $d_s = 2$ they are smooth curves; for $d_s = 3$, smooth surfaces. In every case the set has Hausdorff dimension $d_s - 1 < d_s$, hence zero d_s -dimensional Lebesgue measure (Federer, 1969, Theorem 3.2.22).*

A.2 Proof of Lemma 3.5 (Measure-Zero Indifference)

Proof. Step 1 (Lebesgue measure zero). By Proposition 2.3, each $\mathcal{K}_{a,a'}(\theta_0)$ is either empty or a C^1 submanifold of dimension $d_s - 1$. A C^1 submanifold of $\mathbb{R}^{d_s}$ with dimension strictly less than d_s has d_s -dimensional Lebesgue measure zero (Federer, 1969, Theorem 3.2.22). Since $\mathcal{K}(\theta_0)$ is a finite union of such sets, $\lambda_{d_s}(\mathcal{K}(\theta_0)) = 0$.

Step 2 (Stationary probability zero). By Assumption 4, the stationary distribution μ_{θ_0} admits a bounded density with respect to λ_{d_s} . Absolute continuity and Step 1 give $\mu_{\theta_0}(\mathcal{K}(\theta_0)) = \int_{\mathcal{K}(\theta_0)} m_{\theta_0}(s) ds = 0$. $\square$

A.3 Local uniqueness of maximisers

Lemma A.3. *Under Assumption 5, for each $s \notin \mathcal{K}(\theta_0)$ there exist neighbourhoods $\mathcal{U}_s \ni s$ and $\mathcal{V}_s \ni \theta_0$ such that the optimal action correspondence $\Gamma_\theta(s')$ is a singleton for all $(s', \theta) \in \mathcal{U}_s \times \mathcal{V}_s$, with unique element $a^*(s'; \theta) = a^*(s; \theta_0)$.*

Proof. Since $s \notin \mathcal{K}(\theta_0)$, the maximiser $a^* = a^*(s; \theta_0)$ is unique with gap $\delta := \min_{a \neq a^*} [Q_{\theta_0}(s, a^*; V_{\theta_0}) - Q_{\theta_0}(s, a; V_{\theta_0})] > 0$. By joint continuity of $Q_\theta(s', a; V_\theta)$ in (s', θ) (Assumption 5 and the contraction mapping theorem), there exist $\mathcal{U}_s, \mathcal{V}_s$ such that $Q_\theta(s', a^*; V_\theta) - Q_\theta(s', a; V_\theta) > \delta/2$ for all $(s', \theta) \in \mathcal{U}_s \times \mathcal{V}_s$ and $a \neq a^*$. $\square$

A.4 Proof of Proposition 3.2 (Directional Bellman Derivative)

Proof. We apply the implicit-function theorem for parameterised contractions in Bonnans and Shapiro (2000, Chapter 6, Theorem 6.3.2) in the Banach space $X = (B(\mathcal{S}), \|\cdot\|_\infty)$.

Identification. $\Phi(V, \theta)(s) := (T_\theta V)(s) = \max_{a \in \mathcal{A}} Q_\theta(s, a; V)$; $x_0 = V_{\theta_0}$.

Condition (i): Contraction. For $V_1, V_2 \in B(\mathcal{S})$: $\|T_\theta V_1 - T_\theta V_2\|_\infty \leq \beta \|V_1 - V_2\|_\infty$ (using $|\max_a f_a - \max_a g_a| \leq \max_a |f_a - g_a|$ and $|Q(V_1) - Q(V_2)| \leq \beta \|V_1 - V_2\|_\infty$). The contraction modulus $\beta < 1$ is uniform in θ .

Condition (ii): Hadamard directional differentiability of Φ in θ . For each s , $\max_{a \in \mathcal{A}} Q_\theta(s, a; V)$ is Hadamard directionally differentiable in θ by Proposition C.1 (Appendix C.1), with

$$D_\theta \Phi(V_{\theta_0}, \theta_0)(s)[h] = \max_{a \in \Gamma_{\theta_0}(s)} \nabla_\theta u(s, a; \theta_0)^\top h.$$

Uniformity in s . Define $\varphi_a(s, t) := t^{-1}[Q_{\theta_0+th}(s, a; V_{\theta_0}) - Q_{\theta_0}(s, a; V_{\theta_0})] - \nabla_\theta u(s, a; \theta_0)^\top h$. By the mean-value theorem and uniform continuity of $\nabla_\theta u$ on $\mathcal{S} \times \mathcal{A} \times \Theta$ (compact):

$$\sup_s \sup_a |\varphi_a(s, t)| \leq \omega(|t| \cdot \|h\|) \rightarrow 0 \quad \text{as } t \rightarrow 0,$$

where ω is the modulus of continuity of $\nabla_\theta u$.

For each s , let $a_t^*(s)$ denote any argmax of $Q_{\theta_0+th}(s, \cdot; V_{\theta_0})$ and $a^*(s) \in \Gamma_{\theta_0}(s)$. Then:

$$\begin{aligned} & \left| \frac{(T_{\theta_0+th} V_{\theta_0})(s) - (T_{\theta_0} V_{\theta_0})(s)}{t} - \max_{a \in \Gamma_{\theta_0}(s)} \nabla_\theta u(s, a; \theta_0)^\top h \right| \\ & \leq \left| \max_a [Q_{\theta_0}(s, a; V_{\theta_0}) + t \nabla_\theta u(s, a; \theta_0)^\top h + t \varphi_a(s, t)] \right. \\ & \quad \left. - \max_a Q_{\theta_0}(s, a; V_{\theta_0}) - t \max_{a \in \Gamma_{\theta_0}(s)} \nabla_\theta u(s, a; \theta_0)^\top h \right| \cdot |t|^{-1} + \omega(|t| \|h\|). \end{aligned}$$

By Proposition C.1 applied to $f_a = Q_{\theta_0}(s, a; V_{\theta_0}) + t \nabla_\theta u_a^\top h$ with perturbation $t \varphi_a$, the first term is bounded by $\max_a |\varphi_a(s, t)| \leq \omega(|t| \|h\|)$. Therefore

$$\|t^{-1}[T_{\theta_0+th} V_{\theta_0} - T_{\theta_0} V_{\theta_0}] - D_\theta \Phi(\cdot)[h]\|_\infty \leq 2\omega(|t| \|h\|) \rightarrow 0.$$

Application of Bonnans-Shapiro Theorem 6.3.2. Both conditions verified. The directional derivative W_h is the unique solution in $B(\mathcal{S})$ of $W_h = D_V \Phi(V_{\theta_0}, \theta_0)[W_h] + D_\theta \Phi(V_{\theta_0}, \theta_0)[h]$.

On $\mathcal{S} \setminus \mathcal{K}(\theta_0)$: $D_\theta \Phi(s)[h] = \nabla_\theta u(s, a^*; \theta_0)^\top h$ and $D_V \Phi(s)[W] = \beta \int W(s') F(ds'|s, a^*)$, giving (10).

On $\mathcal{K}(\theta_0)$: $D_\theta \Phi(s)[h] = \max_{a \in \Gamma_{\theta_0}(s)} \nabla_\theta u(s, a; \theta_0)^\top h$, giving (11).

Linearity. On $\mathcal{S} \setminus \mathcal{K}(\theta_0)$, $D_\theta \Phi(s)[h]$ is linear in h (unique argmax), so W_h is linear in h on this set. Since $\mu_{\theta_0}(\mathcal{K}(\theta_0)) = 0$ (Lemma 3.5), W_h depends linearly on h in $L^p(\mu_{\theta_0})$ for any $p \geq 1$. $\square$

Example A.4 (Binary action, one-dimensional state). Let $\mathcal{S} = [0, 1]$, $\mathcal{A} = \{0, 1\}$, $u(s, 0; \theta) = 0$, $u(s, 1; \theta) = s - \theta$ (with $\theta \in [0, 1]$), $F = \text{Unif}[0, 1]$ (independent of (s, a)), and $\beta = 1/2$.

Value function. $Q_\theta(s, 0; V) = \beta \bar{V}$ and $Q_\theta(s, 1; V) = (s - \theta) + \beta \bar{V}$, where $\bar{V} = \int_0^1 V(s') ds'$. The hard-max policy is $a^*(s; \theta) = \mathbf{1}\{s > \theta\}$. Solving for $\bar{V}_\theta$: $\bar{V}_\theta = (1 - \theta)^2/2 + \beta \bar{V}_\theta$, hence $\bar{V}_\theta = (1 - \theta)^2$ and $V_\theta(s) = \max(0, s - \theta) + (1 - \theta)^2/2$.

Indifference set. $\mathcal{K}(\theta) = \{\theta\}$, a single point with Lebesgue measure zero.

Directional derivative. For $h \in \mathbb{R}$ and $s \neq \theta_0$:

$$W_h(s) = \begin{cases} \beta \bar{W}_h & \text{if } s < \theta_0, \\ -h + \beta \bar{W}_h & \text{if } s > \theta_0, \end{cases}$$

where $\bar{W}_h = -(1 - \theta_0)h/(1 - \beta) = -2(1 - \theta_0)h$. Therefore $W_h(s) = -h \cdot \mathbf{1}\{s > \theta_0\} - (1 - \theta_0)h$ for $s \neq \theta_0$. At $s = \theta_0$ (the kink): $W_h(\theta_0) = \max(0, -h) + \beta \bar{W}_h$, illustrating the non-linear dependence on h at the indifference point.

A.5 Differentiability of population criteria

The following result establishes that integrated functionals of the form $M(\theta) := \mathbb{E}_{P_0}[g(S, A, V_\theta, \theta)]$ are differentiable at θ_0 despite pointwise non-smoothness of V_θ . This underpins the score and Hessian calculations in Section 6.

Theorem A.5 (Differentiability of population criteria). Let $g : \mathcal{S} \times \mathcal{A} \times B(\mathcal{S}) \times \Theta \rightarrow \mathbb{R}^m$ be measurable, with $(V, \theta) \mapsto g(s, a, V, \theta)$ Fréchet differentiable on a neighbourhood of (V_{θ_0}, θ_0) for all $s \notin \mathcal{K}(\theta_0)$, and with derivatives dominated by an integrable envelope. Under Assumptions 5–1 and Lemma 3.5, the map $\theta \mapsto M(\theta)$ is directionally differentiable at θ_0 :

$$\nabla_\theta M(\theta_0)^\top h = \mathbb{E}_{P_0}[\nabla_\theta g(S, A, V_{\theta_0}, \theta_0)^\top h + D_V g(S, A, V_{\theta_0}, \theta_0)[W_h]], \quad (\text{A.1})$$

where W_h is the directional derivative from Proposition 3.2.

Proof. Write the difference quotient as $t^{-1}[M(\theta_0 + th) - M(\theta_0)] = I_1(t) + I_2(t)$, where

$$I_1(t) := \mathbb{E}_{P_0}[t^{-1}\{g(S, A, V_{\theta_0}, \theta_0 + th) - g(S, A, V_{\theta_0}, \theta_0)\}], \quad (\text{A.2})$$

$$I_2(t) := \mathbb{E}_{P_0}[t^{-1}\{g(S, A, V_{\theta_0+th}, \theta_0 + th) - g(S, A, V_{\theta_0}, \theta_0 + th)\}]. \quad (\text{A.3})$$

Term I_1 (direct θ -derivative). For $S \notin \mathcal{K}(\theta_0)$, the integrand converges pointwise to $\nabla_\theta g(S, A, V_{\theta_0}, \theta_0)^\top h$ by Fréchet differentiability of g in θ . Since $\mu_{\theta_0}(\mathcal{K}(\theta_0)) = 0$ (Lemma 3.5), this convergence holds P_0 -a.s. The envelope condition and dominated convergence give $I_1(t) \rightarrow \mathbb{E}_{P_0}[\nabla_\theta g^\top h]$.

Term I_2 (indirect V -derivative). Expand g in V around V_{θ_0} via Fréchet differentiability:

$$g(S, A, V_{\theta_0+th}, \theta_0 + th) - g(S, A, V_{\theta_0}, \theta_0 + th) = D_V g(S, A, V_{\theta_0}, \theta_0 + th)[V_{\theta_0+th} - V_{\theta_0}] + \text{rem}_t,$$

where $\|\text{rem}_t\| \leq \epsilon_t \|V_{\theta_0+th} - V_{\theta_0}\|_\infty = o(t)$ by Fréchet differentiability and the contraction bound $\|V_{\theta_0+th} - V_{\theta_0}\|_\infty = O(t)$. Dividing by t and writing $D_t := t^{-1}(V_{\theta_0+th} - V_{\theta_0})$:

$$t^{-1}\{\dots\} = D_V g(S, A, V_{\theta_0}, \theta_0 + th)[D_t] + o(1).$$

By Proposition 3.2, $D_t \rightarrow W_h$ in $L^1(\mu_{\theta_0})$. Continuity of $D_V g$ in θ and the envelope condition give $I_2(t) \rightarrow \mathbb{E}_{P_0}[D_V g(S, A, V_{\theta_0}, \theta_0)[W_h]]$.

Combining the two terms yields (A.1). $\square$

Remark A.6. The gradient formula (A.1) involves no integration over $\mathcal{K}(\theta_0)$: all derivatives are evaluated along the unique optimal policy $a^*(s; \theta_0)$, which is well-defined μ_{θ_0} -a.s. by Lemma 3.5.

Example A.7 (Max operator in $\mathbb{R}^2$). Let $m(x_1, x_2) = \max\{x_1, x_2\}$. The indifference set is $\mathcal{K} = \{(x, x) : x \in \mathbb{R}\}$, a 1-dimensional submanifold (codimension 1).

Away from $\mathcal{K}$: at $(2, 1)$ the maximiser is $a^* = 1$, and the directional derivative in direction $h = (h_1, h_2)$ is h_1 by Lemma A.3.

On $\mathcal{K}$: at $(1, 1)$ with transversal direction $h = (1, 0)$, $m(1+t, 1) = 1+t$, so the derivative is 1. With tangent direction $h = (1, 1)$, $m(1+t, 1+t) = 1+t$, so the derivative is $\max\{h_1, h_2\} = 1$.

For (X_1, X_2) with a smooth density, the integrated criterion $M = \mathbb{E}[m(X_1, X_2)]$ is smooth as a functional of the distribution because $\mathbb{P}((X_1, X_2) \in \mathcal{K}) = 0$ —precisely the MZI phenomenon at work.

A.6 Proof of Theorem 3.8 (Existence of Limiting Quantities)

Proof. Part (i): Existence of H_0 .

For $\tau > 0$, $M(\theta, \tau)$ is smooth in θ (Assumption 5(iv)), so $\nabla^2 M(\theta_0, \tau)$ is well-defined. We show $\tau \mapsto \nabla^2 M(\theta_0, \tau)$ is Cauchy. For $0 < \tau' < \tau$:

$$\|\nabla^2 M(\theta_0, \tau) - \nabla^2 M(\theta_0, \tau')\| \leq \|\nabla^2 M(\theta_0, \tau) - H_\infty\| + \|\nabla^2 M(\theta_0, \tau') - H_\infty\|,$$

where H_∞ is any candidate limit. By Corollary C.3 (Appendix C.2), $\|\nabla^2 M(\theta_0, \tau) - \nabla^2 M(\theta_0, \tau')\| \leq C'''(\tau^\kappa + \tau'^\kappa)[\log(1/\min)]^\kappa$. For $\kappa \geq 2$ both terms vanish, so the sequence is Cauchy.

Negative semidefiniteness: for each small τ , θ_0 maximises $M(\cdot, \tau)$ on Θ by Proposition 4.7, so $\nabla^2 M(\theta_0, \tau) \preceq 0$. The limit of negative semidefinite matrices is negative semidefinite.

Part (ii): Existence of Ω .

Write $\Omega(\tau) = \mathbb{E}[\psi_\tau \psi_\tau^\top] - \mathbb{E}[\psi_\tau] \mathbb{E}[\psi_\tau]^\top$. The second term: $\|\mathbb{E}[\psi_\tau]\| = \|\nabla_\theta M(\theta_0, \tau)\| = O(\tau^\kappa [\log]^\kappa) \rightarrow 0$, so $\mathbb{E}[\psi_\tau] \mathbb{E}[\psi_\tau]^\top \rightarrow 0$.

For the first term, under the correct action $a = a^*(s; \theta_0)$ (which occurs P_0 -a.s.):

$$\psi_\tau(z; \theta_0) = \frac{1}{\tau} \sum_{a' \neq a^*} p_\tau(a'|s) [\nabla_\theta Q_{a^*}(s) - \nabla_\theta Q_{a'}(s)].$$

Using $p_\tau(a'|s) \leq \exp(-\Delta_{a^*, a'}(s)/\tau)$ (where $\Delta = Q_{a^*} - Q_{a'}$):

$$\|\psi_\tau(z; \theta_0)\| \leq \frac{C|\mathcal{A}|}{\tau} \exp(-\Delta(s)/\tau), \quad (\text{A.4})$$

with $\Delta(s) := \min_{a' \neq a^*} \Delta_{a^*, a'}(s; \theta_0)$.

Second moment.

$$\mathbb{E}[\|\psi_\tau\|^2] \leq \frac{C^2 |\mathcal{A}|^2}{\tau^2} \int_0^\infty e^{-2\delta/\tau} f_\Delta(\delta) d\delta,$$

where f_Δ is the density of $\Delta(S)$ under μ_{θ_0} . The margin condition (22) implies $f_\Delta(\delta) \leq C_\kappa \delta^{\kappa-1}$ for small δ . Substituting $x = \delta/\tau$:

$$\int_0^\infty e^{-2\delta/\tau} \delta^{\kappa-1} d\delta = \tau^\kappa \int_0^\infty e^{-2x} x^{\kappa-1} dx = \tau^\kappa \frac{\Gamma(\kappa)}{2^\kappa}.$$

Therefore $\mathbb{E}[\|\psi_\tau\|^2] \leq C' \tau^{\kappa-2}$. For $\kappa = 2$: $O(1)$, bounded. For $\kappa > 2$: $O(\tau^{\kappa-2}) \rightarrow 0$.

Convergence of $\tau \mapsto \mathbb{E}[\psi_\tau \psi_\tau^\top]$. By dominated convergence on $\{\Delta > \delta_0\}$ (where ψ_τ vanishes exponentially) and the Vitali convergence theorem on $\{\Delta \leq \delta_0\}$ (using the $L^{2+\delta}$ bound from Appendix B.3), the map $\tau \mapsto \mathbb{E}[\psi_\tau \psi_\tau^\top]$ has a limit as $\tau \downarrow 0$.

Part (iii): Immediate from Proposition 4.4. □

A.7 Proof of Proposition 4.7 (Annealing Neutrality)

Proof. The proof in Section 4.4 is complete. We verify the key claim in Step 2: positivity of $\eta(\varepsilon)$.

Lower semicontinuity. Define $g(\theta) := \mu_{\theta_0}(\{s : a^*(s; \theta) \neq a^*(s; \theta_0)\})$. For any sequence $\theta_n \rightarrow \theta$, $\mathbf{1}\{a^*(s; \theta_n) \neq a^*(s; \theta_0)\} \rightarrow \mathbf{1}\{a^*(s; \theta) \neq a^*(s; \theta_0)\}$ for μ_{θ_0} -a.e. $s \notin \mathcal{K}(\theta) \cup \mathcal{K}(\theta_0)$. By Fatou's lemma, $g(\theta) \leq \liminf g(\theta_n)$, so g is lower semicontinuous.

Positivity on $\Theta \setminus B(\theta_0, \varepsilon)$. By Assumption 3, $g(\theta) > 0$ for all $\theta \neq \theta_0$. Since g is lower semicontinuous and $\Theta \setminus B(\theta_0, \varepsilon)$ is compact, the infimum $\eta(\varepsilon) = \inf_{\theta \notin B(\theta_0, \varepsilon)} g(\theta)$ is attained at some $\theta^* \neq \theta_0$, where $g(\theta^*) > 0$. $\square$

B Proofs of Asymptotic Theory

B.1 Proof of Theorem 6.2 (Consistency)

Proof. We verify the three conditions of Newey and McFadden (1994, Theorem 2.1) for the criterion $\hat{Q}_n(\theta) := \hat{L}_n(\theta, \tau_n; \hat{V}_n)$ and the limiting criterion $Q_0(\theta) := M(\theta, 0)$.

Condition 1 (Identification). $\theta_0 = \arg \max_{\theta} M(\theta, 0)$ is unique by Assumption 3.

Condition 2 (Compactness). Θ is compact by Assumption 5(iii).

Condition 3 (Uniform convergence). Decompose

$$\sup_{\theta} |\hat{Q}_n(\theta) - Q_0(\theta)| \leq R_1 + R_2 + R_3, \quad (\text{B.1})$$

where

$$R_1 := \sup_{\theta} |\hat{L}_n(\theta, \tau_n; \hat{V}_n) - \hat{L}_n(\theta, \tau_n; V_{\theta, \tau_n})| \leq \frac{2\beta}{\tau_n} \varepsilon_n, \quad (\text{B.2})$$

$$R_2 := \sup_{\theta} |\hat{L}_n(\theta, \tau_n; V_{\theta, \tau_n}) - M(\theta, \tau_n)|, \quad (\text{B.3})$$

$$R_3 := \sup_{\theta} |M(\theta, \tau_n) - M(\theta, 0)|. \quad (\text{B.4})$$

R_1 : Lemma 4.5 and (R3) give $\varepsilon_n/\tau_n \rightarrow 0$, so $R_1 \rightarrow 0$.

R_2 : Under Assumption 4(a), the class $\{\ell(\cdot; \theta, \tau_n; V_{\theta, \tau_n}) : \theta \in \Theta\}$ is a Lipschitz image of Θ (compact) for each $\tau_n > 0$, hence Glivenko–Cantelli (van der Vaart, 1998, Theorem 19.4). Under Assumption 4(b), geometric ergodicity and Andrews (1992, Theorem 3.1) give $R_2 \xrightarrow{p} 0$.

R_3 : The uniform smoothing convergence in Appendix A.7, Step 2, gives $R_3 \rightarrow 0$.

All three conditions hold, so $\hat{\theta}_n \xrightarrow{p} \theta_0$. $\square$

B.2 Proof of Theorem 6.3 (Asymptotic Normality)

Proof. Step 1 (Mean-value expansion). Since $\ell(z; \theta, \tau; V)$ is twice continuously differentiable in θ for $\tau > 0$, the first-order condition $\nabla_{\theta} \hat{L}_n(\hat{\theta}_n, \tau_n; \hat{V}_n) = 0$ and a mean-value expansion around θ_0 give (32), with $\|\tilde{\theta}_n - \theta_0\| \leq \|\hat{\theta}_n - \theta_0\| = o_p(1)$ by consistency.

Step 2 (Score decomposition). Add and subtract $\sqrt{n} \nabla_{\theta} \hat{L}_n(\theta_0, \tau_n; V_{\theta_0, \tau_n})$ and $\sqrt{n} \mathbb{E}[\psi_{\tau_n}(Z; \theta_0)]$:

$$\sqrt{n} \nabla_{\theta} \hat{L}_n(\theta_0, \tau_n; \hat{V}_n) = \underbrace{\frac{1}{\sqrt{n}} \sum_{i=1}^n [\psi_{\tau_n}(z_i; \theta_0) - \mathbb{E}[\psi_{\tau_n}]]}_{=: S_n} + \underbrace{\sqrt{n} [\bar{\psi}_{\tau_n}(\hat{V}_n) - \bar{\psi}_{\tau_n}(V_{\theta_0, \tau_n})]}_{=: A_n} + \underbrace{\sqrt{n} \mathbb{E}[\psi_{\tau_n}]}_{=: B_n}.$$

Step 3 ($A_n = o_p(1)$: *approximation bias*). The Jacobian $\nabla_V \nabla_\theta \ell$ of the score with respect to V satisfies

$$\nabla_V \nabla_\theta \ell(z; \theta_0, \tau_n; V) = \frac{1}{\tau_n} \nabla_\theta \left[\frac{\partial Q_a}{\partial V} - \sum_{a'} p_{a'} \frac{\partial Q_{a'}}{\partial V} \right],$$

where $\partial Q_a / \partial V = \beta F(\cdot | s, a)$, so $\|\nabla_V \nabla_\theta \ell\| \leq C\beta/\tau_n$ for a constant C depending on $\|\nabla_\theta u\|_\infty$. By the mean-value theorem in V : $\|A_n\| \leq \sqrt{n} \cdot (C\beta/\tau_n) \cdot \varepsilon_n = O(\sqrt{n} \varepsilon_n / \tau_n) \rightarrow 0$ under (R3).

Step 4 ($B_n = o(1)$: *smoothing bias*). Since $\nabla_\theta M(\theta_0, 0) = 0$ by optimality of θ_0 , $\mathbb{E}[\psi_{\tau_n}] = \nabla_\theta M(\theta_0, \tau_n) - \nabla_\theta M(\theta_0, 0)$. Differentiating the smoothing bias bound (Proposition 4.4): $\|\mathbb{E}[\psi_{\tau_n}]\| \leq C' \tau_n^\kappa [\log(1/\tau_n)]^\kappa$. Therefore $\|B_n\| = \sqrt{n} \cdot O(\tau_n^\kappa [\log(1/\tau_n)]^\kappa) \rightarrow 0$ under (R4).

Step 5 ($S_n \Rightarrow N(0, \Omega)$: *CLT*).

Variance convergence. By Lemma 4.6, $\text{Var}(\psi_{\tau_n}) \rightarrow \Omega$ as $n \rightarrow \infty$.

Lyapunov condition. By Lemma B.1 (Appendix B.3), there exist $\delta > 0$ and $C > 0$ such that $\mathbb{E}[\|\psi_{\tau_n}\|^{2+\delta}] \leq C$ for all n . The Lyapunov sum is

$$\sum_{i=1}^n \mathbb{E} \left[\left\| \frac{\psi_{\tau_n}(z_i)}{\sqrt{n}} \right\|^{2+\delta} \right] = \frac{\mathbb{E}[\|\psi_{\tau_n}\|^{2+\delta}]}{n^{\delta/2}} \leq \frac{C}{n^{\delta/2}} \rightarrow 0.$$

Application. Under Assumption 4(a), the multivariate Lyapunov CLT (Billingsley, 1995, Theorem 27.3) gives $S_n \Rightarrow N(0, \Omega)$. Under Assumption 4(b), the CLT for geometrically ergodic Markov chains (Jones and Hobert, 2001, Theorem 4) applies; the limiting variance becomes $\Omega_{LR} = \sum_{k=-\infty}^{\infty} \text{Cov}(\psi_0(z_0), \psi_0(z_k))$.

Step 6 (*Hessian convergence*). Decompose $\hat{H}_n - H_0 = H_1 + H_2 + H_3 + H_4$, where

$$\begin{aligned} H_1 &:= \nabla^2 \hat{L}_n(\tilde{\theta}_n, \tau_n; \hat{V}_n) - \nabla^2 \hat{L}_n(\tilde{\theta}_n, \tau_n; V_{\theta_0, \tau_n}), \\ H_2 &:= \nabla^2 \hat{L}_n(\tilde{\theta}_n, \tau_n; V_{\theta_0, \tau_n}) - \nabla^2 \hat{L}_n(\theta_0, \tau_n; V_{\theta_0, \tau_n}), \\ H_3 &:= \nabla^2 \hat{L}_n(\theta_0, \tau_n; V_{\theta_0, \tau_n}) - \mathbb{E}[\nabla^2 \ell(Z; \theta_0, \tau_n; V_{\theta_0, \tau_n})], \\ H_4 &:= \mathbb{E}[\nabla^2 \ell(Z; \theta_0, \tau_n; V_{\theta_0, \tau_n})] - H_0. \end{aligned} \tag{B.5}$$

$H_1 = o_p(1)$. The softmax Hessian depends on V through $Q_\theta(s, a; V)$ and $p_{\theta, \tau}(a|s; V)$, with $\|H_1\| \leq C\beta\varepsilon_n/\tau_n^2$. Since $\varepsilon_n = O(n^{-\alpha})$ and $\tau_n = n^{-\beta}$, $\varepsilon_n/\tau_n^2 = O(n^{-\alpha+2\beta})$. Under (R3) ($\alpha > \beta + 1/2$) and $\beta < 1/2$, we have $\alpha > 2\beta$, so $H_1 = o_p(1)$.

$H_2 = o_p(1)$. By $\tilde{\theta}_n \xrightarrow{p} \theta_0$ and stochastic equicontinuity of the empirical Hessian process (Andrews, 1992, Theorem 2).

$H_3 = o_p(1)$. Lemma B.4 (Appendix B.4).

$H_4 = o(1)$. Corollary C.3: $\|H_4\| \leq C'' \tau_n^\kappa [\log(1/\tau_n)]^\kappa \rightarrow 0$.

Combining Steps 1–6. From (32):

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -\hat{H}_n^{-1}[S_n + A_n + B_n] = -H_0^{-1}S_n + o_p(1) \xrightarrow{d} N(0, H_0^{-1}\Omega H_0^{-1}),$$

using $\hat{H}_n \rightarrow H_0$ (invertible by Assumption 8(ii)), $A_n = o_p(1)$, $B_n = o(1)$, and $S_n \Rightarrow N(0, \Omega)$. $\square$

B.3 Verification of the Lyapunov Condition

Lemma B.1. *Under Assumptions 5–1 and the margin condition (22) with $\kappa > 2$, there exists $\delta \in (0, \kappa - 2]$ such that $\sup_{\tau \in (0, 1]} \mathbb{E}[\|\psi_\tau(Z; \theta_0)\|^{2+\delta}] < \infty$.*

Proof. The annealed score under the correct action $a = a^*(s; \theta_0)$ satisfies (A.4):

$$\|\psi_\tau(z; \theta_0)\| \leq \frac{C|\mathcal{A}|}{\tau} \exp(-\Delta(s)/\tau).$$

Moment bound.

$$\mathbb{E}[\|\psi_\tau\|^{2+\delta}] \leq \frac{(C|\mathcal{A}|)^{2+\delta}}{\tau^{2+\delta}} \int_0^\infty e^{-(2+\delta)\delta'/\tau} f_\Delta(\delta') d\delta'.$$

Using $f_\Delta(\delta') \leq C_\kappa \kappa (\delta')^{\kappa-1}$ and the substitution $x = \delta'/\tau$:

$$\int_0^\infty e^{-(2+\delta)\delta'/\tau} (\delta')^{\kappa-1} d\delta' = \tau^\kappa \int_0^\infty e^{-(2+\delta)x} x^{\kappa-1} dx = \tau^\kappa \frac{\Gamma(\kappa)}{(2+\delta)^\kappa}.$$

Therefore:

$$\mathbb{E}[\|\psi_\tau\|^{2+\delta}] \leq C'' \tau^{\kappa-2-\delta}. \quad (\text{B.6})$$

This is bounded uniformly in $\tau \in (0, 1]$ if and only if $\kappa - 2 - \delta \geq 0$, i.e., $\delta \leq \kappa - 2$.

For $\kappa > 2$: choose $\delta \in (0, \kappa - 2]$ (e.g. $\delta = (\kappa - 2)/2$) and $\sup_\tau \mathbb{E}[\|\psi_\tau\|^{2+\delta}] < \infty$.

For $\kappa = 2$: bound (B.6) gives $\mathbb{E}[\|\psi_\tau\|^{2+\delta}] \leq C'' \tau^{-\delta}$, which diverges for any $\delta > 0$. The Lindeberg condition (which requires only second moments) also fails with the crude bound: the truncated second moment $n \cdot (C^2/\tau_n^2) \cdot \mu_{\theta_0}(\Delta \leq \tau_n \log(\dots)) \leq n \cdot C_2 [\log(\dots)]^2 = O((\log n)^2)$ does not vanish. We impose $\kappa > 2$ in Assumption 8(iii–iv). $\square$

Remark B.2 (Sharpness of $\kappa > 2$). *The condition $\kappa > 2$ is sharp for the margin-integral technique. Alternative CLT approaches—martingale CLTs under ergodicity, or truncation-based CLTs—may extend the result to $\kappa \geq 2$ or $\kappa > 1$.*

Example B.3 (Score variance for the uniform-state model). *Continuing Example A.4: $\mathcal{S} = [0, 1]$, $\mathcal{A} = \{0, 1\}$, $u(s, 1; \theta) = s - \theta$, $u(s, 0; \theta) = 0$, $F = \text{Unif}[0, 1]$, $\beta = 1/2$, $\theta_0 = 1/2$.*

The value-function gap is $\Delta(s) = |s - 1/2|$ and the invariant distribution is $\mu = \text{Unif}[0, 1]$, giving $\mu(\Delta \leq \delta) = 2\delta$ and $\kappa = 1$.

The annealed score at θ_0 with $a = a^(s) = \mathbf{1}\{s > 1/2\}$ is:*

$$\psi_\tau(s; \theta_0) = \begin{cases} -\frac{1}{\tau} \cdot \frac{\exp(-(s - 1/2)/\tau)}{1 + \exp(-(s - 1/2)/\tau)} & \text{if } s > 1/2, \\ \frac{1}{\tau} \cdot \frac{1}{1 + \exp((s - 1/2)/\tau)} & \text{if } s < 1/2. \end{cases}$$

Second moment.

$$\mathbb{E}[\psi_\tau^2] = 2 \int_0^{1/2} \frac{1}{\tau^2} \cdot \frac{e^{-2\delta/\tau}}{(1 + e^{-\delta/\tau})^4} d\delta = \frac{2}{\tau} \int_0^{1/(2\tau)} \frac{e^{-2x}}{(1 + e^{-x})^4} dx.$$

As $\tau \rightarrow 0$ the upper limit diverges and $\int_0^\infty e^{-2x}/(1 + e^{-x})^4 dx = 1/6$, so $\mathbb{E}[\psi_\tau^2] \sim 1/(3\tau) \rightarrow \infty$.

For this model with $\kappa = 1$, the score variance diverges as $\tau \rightarrow 0$. The standard $\sqrt{n}$ CLT applied to $n^{-1/2} \sum_i \psi_{\tau_n}(z_i)$ produces a degenerate limit: the variance grows like $n \cdot \tau_n^{-1} = n^{1+\beta}$. A different normalisation—e.g. $n^{(1-\beta)/2}$ —would be needed for a non-degenerate limit, illustrating why $\kappa \geq 2$ is required in the formal theory.

B.4 Uniform Law of Large Numbers for the Hessian

Lemma B.4. *Under the conditions of Theorem 6.3, for any sequence $(\tilde{\theta}_n, \tau_n) \rightarrow (\theta_0, 0)$:*

$$\frac{1}{n} \sum_{i=1}^n \nabla_{\theta\theta}^2 \ell(z_i; \tilde{\theta}_n, \tau_n; V_{\tilde{\theta}_n, \tau_n}) \xrightarrow{p} H_0.$$

Proof. Define $\phi(z; \theta, \tau) := \nabla_{\theta\theta}^2 \ell(z; \theta, \tau; V_{\theta, \tau})$ and $\bar{\phi}(\theta, \tau) := \mathbb{E}[\phi(Z; \theta, \tau)]$.

Step 1 (Pointwise limit). $\bar{\phi}(\theta_0, \tau_n) \rightarrow H_0$ by Corollary C.3 (Appendix C.2).

Step 2 (Concentration). For each n , $\phi(\cdot; \tilde{\theta}_n, \tau_n)$ is a measurable function bounded by C/τ_n^2 . By the triangular-array LLN:

$$\frac{1}{n} \sum_i \phi(z_i; \tilde{\theta}_n, \tau_n) - \bar{\phi}(\tilde{\theta}_n, \tau_n) \xrightarrow{p} 0,$$

provided the variance vanishes. By the same margin-integral calculation as (B.6):

$$\text{Var}(\phi) \leq \mathbb{E}[\phi^2] \leq C''' \tau_n^{\kappa-4}.$$

For $\kappa \geq 2$: $\text{Var}(\phi)/n = O(\tau_n^{\kappa-4}/n) = O(n^{(4-\kappa)\beta-1})$. Since $\beta < 1/2$ and $\kappa \geq 2$: $(4-\kappa)\beta \leq 2\beta < 1$, so the exponent is negative and $\text{Var}(\phi)/n \rightarrow 0$.

Step 3 (Continuity of the limit). $\bar{\phi}(\theta, \tau)$ is continuous in (θ, τ) for $\tau > 0$, and $\bar{\phi}(\theta_0, \tau_n) \rightarrow H_0$ by Step 1. Since $\tilde{\theta}_n \rightarrow \theta_0$:

$$|\bar{\phi}(\tilde{\theta}_n, \tau_n) - \bar{\phi}(\theta_0, \tau_n)| \leq L(\tau_n) \|\tilde{\theta}_n - \theta_0\|,$$

where $L(\tau_n) = O(\tau_n^{-2}) = O(n^{2\beta})$ is the Lipschitz constant. With $\|\hat{\theta}_n - \theta_0\| = O_p(n^{-1/2})$, the product vanishes when $2\beta < 1/2$, i.e. $\beta < 1/4$. For $\beta \in [1/4, 1/2)$, a refined argument using the score structure near the threshold (analogous to Appendix B.3) is needed; see Andrews (1992, Theorem 2) for the stochastic equicontinuity framework.

For $\beta < 1/4$ (within the feasible range): $n^{-1} \sum_i \phi(z_i; \tilde{\theta}_n, \tau_n) \rightarrow H_0$ in probability. $\square$

C Auxiliary Results

C.1 Hadamard Directional Differentiability of the Max Operator

Proposition C.1 (Finite-dimensional max). *Let $f_1, \dots, f_J : \mathbb{R}^m \rightarrow \mathbb{R}$ be differentiable at x_0 . Define $F(x) = \max_{j=1, \dots, J} f_j(x)$ and $\mathcal{A}^*(x_0) = \{j : f_j(x_0) = F(x_0)\}$. Then F is Hadamard directionally differentiable at x_0 :*

$$F'(x_0; h) = \max_{j \in \mathcal{A}^*(x_0)} \nabla f_j(x_0)^\top h.$$

Proof. This is Bonnans and Shapiro (2000, Proposition 2.47); see also Rockafellar and Wets (1998, Theorem 10.31). Let $t_n \downarrow 0$ and $h_n \rightarrow h$. For $j \in \mathcal{A}^*(x_0)$: $t_n^{-1}[f_j(x_0 + t_n h_n) - f_j(x_0)] \rightarrow \nabla f_j(x_0)^\top h$. For $j \notin \mathcal{A}^*(x_0)$: continuity and the strict gap give $f_j(x_0 + t_n h_n) < F(x_0)$ for n large, so j is not the argmax. Therefore $t_n^{-1}[F(x_0 + t_n h_n) - F(x_0)] \rightarrow \max_{j \in \mathcal{A}^*} \nabla f_j(x_0)^\top h$. $\square$

Remark C.2 (Extension to function spaces). *For the Bellman operator $T_\theta V = \max_a Q_\theta(\cdot, a; V)$ on $B(\mathcal{S})$, Hadamard directional differentiability holds pointwise (at each s) by Proposition C.1, since the max is over the finite set $\mathcal{A}$. Integration against μ_{θ_0} and the MZI property (Lemma 3.5) then give differentiability of the integrated operator. For the fixed point $V_\theta = T_\theta V_\theta$, we use the parameterised-contraction approach (Appendix A.4) rather than directly composing Hadamard differentiability with the fixed-point map, avoiding the need for tangential Hadamard differentiability in the sense of Shapiro (1990).*

C.2 Smoothing Bias Bounds

Proof of Proposition 4.4. Let $a^* = a^*(s; \theta)$ and $\Delta(s) := \min_{a' \neq a^*} [Q_{a^*}(s; V_\theta) - Q_{a'}(s; V_\theta)]$. Fix $\delta > 0$ and decompose:

$$M(\theta, \tau) - M(\theta, 0) = \underbrace{\int_{\{\Delta > \delta\}} [\log p_{\theta, \tau}(a|s; V_{\theta, \tau}) - \log p_{\theta, 0}(a|s; V_\theta)] dP_0}_{=: I_{\text{far}}} + \underbrace{\int_{\{\Delta \leq \delta\}} [\dots] dP_0}_{=: I_{\text{near}}}.$$

Far-from-threshold region. On $\{\Delta > \delta\}$, $a = a^*$ with P_0 -probability one and $\log p_{\theta, 0}(a^*|s) = 0$. Using $\|V_{\theta, \tau} - V_\theta\|_\infty \leq (1 - \beta)^{-1} \tau \log |\mathcal{A}|$ (Lemma 4.1), the effective gap under $V_{\theta, \tau}$ is at least $\Delta - 2\beta(1 - \beta)^{-1} \tau \log |\mathcal{A}| \geq \Delta/2$ for τ small enough relative to δ . Therefore $|\log p_{\theta, \tau}(a^*|s; V_{\theta, \tau})| \leq |\mathcal{A}| \exp(-\Delta/(2\tau))$, giving $|I_{\text{far}}| \leq |\mathcal{A}| \exp(-\delta/(2\tau))$.

Near-threshold region. $|I_{\text{near}}| \leq \log |\mathcal{A}| \cdot \mu_{\theta_0}(\Delta \leq \delta) \leq C_\kappa \log |\mathcal{A}| \cdot \delta^\kappa$ by the margin condition.

Optimisation over δ . Set $\delta = 2\tau \log(1/\tau)$. Then $|I_{\text{far}}| \leq |\mathcal{A}| \tau$ and $|I_{\text{near}}| \leq C_\kappa \log |\mathcal{A}| [2\tau \log(1/\tau)]^\kappa$, so $|M(\theta, \tau) - M(\theta, 0)| \leq C \tau^\kappa [\log(1/\tau)]^\kappa$ with $C = C_\kappa \log |\mathcal{A}| \cdot 2^\kappa + |\mathcal{A}|$. $\square$

Corollary C.3 (Hessian bias bound). *Under the same conditions and $u \in C^3(\Theta)$,*

$$\|\nabla_{\theta\theta}^2 M(\theta_0, \tau) - H_0\| \leq C'' \tau^\kappa [\log(1/\tau)]^\kappa.$$

Proof. Differentiate the bias decomposition twice with respect to θ . Each differentiation introduces factors involving derivatives of u and V_θ , but the exponential decay on $\{\Delta > \delta\}$ and the margin condition on $\{\Delta \leq \delta\}$ provide the same rate. $\square$

C.3 Neural Network Approximation Rates

Proof of Proposition 5.2. The value function V_θ is piecewise smooth: it belongs to $C^r(\mathcal{S} \setminus \mathcal{K}(\theta))$ with $r \geq 2$ (Assumption 6) and has Lipschitz kinks along the $(d_s - 1)$ -dimensional manifold $\mathcal{K}(\theta)$ (Proposition 2.3).

Smooth approximation. By Yarotsky (2017, Theorem 1), a ReLU network with $O(N \log N)$ parameters achieves approximation error $O(N^{-2r/d_s})$ in $\|\cdot\|_\infty$ for C^r functions on $[0, 1]^{d_s}$. With $N = n^{d_s/(2r+d_s)}$ parameters (up to log factors): $\varepsilon_{\text{smooth}} = O(n^{-r/(2r+d_s)})$.

Kink handling. By Petersen and Voigtlaender (2018, Theorem 3.2), the additional cost of approximating piecewise smooth functions with kinks along $(d_s - 1)$ -dimensional manifolds is at most a logarithmic factor: $\varepsilon_{\text{kink}} = O((\log n)^2 \cdot n^{-r/(2r+d_s)})$.

Uniformity in θ . Since V_θ depends continuously on θ (contraction property) and Θ is compact, a covering-number argument gives uniform approximation: cover Θ with $O(\varepsilon^{-d_\theta})$ balls and use the Lipschitz bound $\|V_{\theta_1} - V_{\theta_2}\|_\infty \leq (1 - \beta)^{-1} \|\nabla_\theta u\|_\infty \|\theta_1 - \theta_2\|$ on each ball.

Combining: $\varepsilon_n = \sup_\theta \|\hat{V}_\theta^{(n)} - V_\theta\|_\infty = O(n^{-r/(2r+d_s)} (\log n)^2)$. $\square$

Remark C.4 (Comparison to polynomial sieves). *Polynomial sieves (Kristensen et al., 2021) achieve similar rates for smooth functions but require $O(K^{d_s})$ basis functions of degree K in each dimension. Neural networks with depth $O(\log n)$ and width $O(n^{d_s/(2r+d_s)})$ have $O(n^{d_s/(2r+d_s)} \log n)$ parameters, growing polynomially rather than exponentially in d_s . This is the primary computational advantage for $d_s \geq 4$.*

C.4 Lipschitz Bound for the Annealed Log-Likelihood

Proof of Lemma 4.5. Write

$$\ell(z; \theta, \tau; V_1) - \ell(z; \theta, \tau; V_2) = \frac{1}{\tau} [Q_\theta(s, a; V_1) - Q_\theta(s, a; V_2)] - \frac{1}{\tau} [\text{LSE}(s; V_1) - \text{LSE}(s; V_2)],$$

where $\text{LSE}(s; V) := \tau \log \sum_{a'} \exp[Q_\theta(s, a'; V)/\tau]$.

First term. $|Q_\theta(s, a; V_1) - Q_\theta(s, a; V_2)| = \beta |\int [V_1 - V_2] F(ds' | s, a)| \leq \beta \|V_1 - V_2\|_\infty$.

Second term. The log-sum-exp is nonexpansive: for $x, y \in \mathbb{R}^J$, $|\tau \log \sum_j e^{x_j/\tau} - \tau \log \sum_j e^{y_j/\tau}| \leq \max_j |x_j - y_j|$. Applied with $x_j = Q_\theta(s, j; V_1)$ and $y_j = Q_\theta(s, j; V_2)$: $|\text{LSE}(s; V_1) - \text{LSE}(s; V_2)| \leq \beta \|V_1 - V_2\|_\infty$.

Combined. $|\ell_1 - \ell_2| \leq (2\beta/\tau) \|V_1 - V_2\|_\infty$. The explicit constant 2β (rather than an unspecified C/τ) arises from the two-term decomposition: one factor of β from discounting in the continuation value, and the factor of 2 from the presence of Q_a in both the numerator and the normalisation. $\square$

C.5 Score Convergence Detail

Proof of Lemma 4.6. We show $\Omega(\tau) := \text{Var}(\psi_\tau(Z; \theta_0))$ converges as $\tau \downarrow 0$.

Step 1 (Pointwise convergence). For $s \notin \mathcal{K}(\theta_0)$, $\|\psi_\tau(s; \theta_0)\| \leq (C/\tau) \exp(-\Delta(s)/\tau) \rightarrow 0$ exponentially.

Step 2 (Uniform integrability of $\|\psi_\tau\|^2$). By Lemma B.1 with $\kappa > 2$, $\sup_\tau \mathbb{E}[\|\psi_\tau\|^{2+\delta}] < \infty$ for some $\delta > 0$. The de la Vallée-Poussin criterion gives uniform integrability of $\{\|\psi_\tau\|^2\}_\tau$.

Step 3 (Vitali's theorem). Pointwise convergence (Step 1) plus uniform integrability (Step 2) imply $\mathbb{E}[\psi_\tau \psi_\tau^\top] \rightarrow \Omega_\infty$ for some Ω_∞ . Since $\mathbb{E}[\psi_\tau] \mathbb{E}[\psi_\tau]^\top \rightarrow 0$, $\text{Var}(\psi_\tau) \rightarrow \Omega_\infty$.

For $\kappa = 2$: uniform integrability fails (Lemma B.1), but $\mathbb{E}[\|\psi_\tau\|^2] = O(1)$, so $\Omega(\tau)$ is bounded. Convergence requires an alternative argument (e.g., monotone convergence on suitable decompositions). We impose $\kappa > 2$ for the full result or assume $\Omega(\tau) \rightarrow \Omega$ directly (Assumption 8(i)). $\square$

D Monte Carlo Design and Algorithms for Design 1

This appendix documents the Monte Carlo design and numerical implementation of the grid-based experiment (Design 1), in which ANE is implemented without sieve approximation error ($\varepsilon_n = 0$). As established in Remark 5.5, the grid-based ANE is the *operative* theoretical result: the rate conditions (R1)–(R4) are feasible for any $\kappa > 1$ with $\beta \in (1/(2\kappa), 1/2)$, whereas no finite smoothness r suffices for the neural-network sieve when $d_s \geq 1$.

D.1 Dynamic labour-supply model

The primitives are as in Section 8: state $s_t \in \mathcal{S} = [s_{\min}, s_{\max}]$, actions $a_t \in \{0, 1\}$, utility

$$u(s, a; \theta) = \begin{cases} \frac{(s + wa)^{1-\gamma} - 1}{1-\gamma} - \alpha a, & \gamma \neq 1, \\ \log(s + wa) - \alpha a, & \gamma = 1, \end{cases} \quad (\text{D.1})$$

transition

$$s_{t+1} = \rho s_t + w a_t + \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, \sigma^2), \quad (\text{D.2})$$

and discount factor $\beta \in (0, 1)$. We discretise $\mathcal{S}$ on a uniform grid $G = \{s_1, \dots, s_M\}$ with $s_1 = s_{\min}$, $s_M = s_{\max}$, and constant spacing $\Delta s = (s_{\max} - s_{\min})/(M - 1)$.

For $a \in \{0, 1\}$, the transition from s_i to s_j is represented by a matrix $P_a \in [0, 1]^{M \times M}$ with

$$P_a(i, j) = \Pr\{s_{t+1} = s_j \mid s_t = s_i, a_t = a\},$$

which we approximate by Monte Carlo:

$$P_a(i, j) \approx \frac{1}{N_{\text{sim}}} \sum_{k=1}^{N_{\text{sim}}} \mathbf{1}\{\text{round}_G(\rho s_i + w a + \varepsilon^{(k)}) = s_j\},$$

where $\varepsilon^{(k)} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$, $N_{\text{sim}} = 2,000$, and round_G maps a continuous state to the nearest grid point (with truncation to $[s_{\min}, s_{\max}]$). The seed for P_a is fixed across replications (seed = 123 + a), so the transition matrices are identical for all Monte Carlo draws.

D.2 Annealed Bellman operator and CCPs

On the grid, the choice-specific values at $(\theta, V) \in \Theta \times \mathbb{R}^M$ are

$$Q_\theta(s_i, a; V) = u(s_i, a; \theta) + \beta \sum_{j=1}^M P_a(i, j) V(s_j).$$

The annealed (log-sum-exp) Bellman operator is

$$(T_{\theta, \tau} V)(s_i) = \tau \log\left(\exp(Q_\theta(s_i, 0; V)/\tau) + \exp(Q_\theta(s_i, 1; V)/\tau)\right). \quad (\text{D.3})$$

For numerical stability, we use the identity $\tau \log(\sum_a e^{Q_a/\tau}) = \max_a Q_a + \tau \log(\sum_a e^{(Q_a - \max Q)/\tau})$.

For each $\tau > 0$ and θ , $T_{\theta, \tau}$ is a contraction on $\mathbb{R}^M$ with modulus β and admits a unique fixed point $V_{\theta, \tau}$ satisfying $V_{\theta, \tau} = T_{\theta, \tau} V_{\theta, \tau}$. We iterate until $\|V^{(k+1)} - V^{(k)}\|_\infty < 10^{-6}$ or 2,000 iterations.

The annealed conditional choice probabilities on the grid are

$$p_{\theta, \tau}(1 \mid s_i) = \frac{\exp(Q_\theta(s_i, 1; V_{\theta, \tau})/\tau)}{\exp(Q_\theta(s_i, 0; V_{\theta, \tau})/\tau) + \exp(Q_\theta(s_i, 1; V_{\theta, \tau})/\tau)}.$$

For data $(s_t, a_t)_{t=1}^n$ generated from the hard-max DGP, the annealed log-likelihood at (θ, τ) is

$$\ell_n(\theta, \tau) = \sum_{t=1}^n \log p_{\theta, \tau}(a_t \mid s_t),$$

where $p_{\theta, \tau}(\cdot \mid s)$ at a continuous state s is obtained by *linear interpolation* of the grid CCPs $\{p_{\theta, \tau}(\cdot \mid s_i)\}_{i=1}^M$. The probabilities are clipped to $[10^{-10}, 1 - 10^{-10}]$ before taking logarithms.

D.3 Data-generating process

The DGP simulates from the hard-max policy at θ_0 using a fine grid G_{DGP} with $M_{\text{DGP}} = 200$ points.

Algorithm 4 Data generation for Design 1 and Design 2

Require: True parameter $\theta_0 = (\alpha_0, \gamma_0) = (1.8, 2.0)$; sample size n ; seed r

- 1: Construct grid G_{DGP} on $[0.1, 10.0]$, $M_{\text{DGP}} = 200$ points.
- 2: Construct transition matrices P_0, P_1 on G_{DGP} ($N_{\text{sim}} = 2,000$, seed = $123 + a$).
- 3: Solve V_{θ_0} via annealed value iteration with $\tau = 10^{-9}$ (tolerance 10^{-6}).
- 4: Compute hard-max policy: $a^*(s_i) = \mathbf{1}\{Q_{\theta_0}(s_i, 1; V_{\theta_0}) \geq Q_{\theta_0}(s_i, 0; V_{\theta_0})\}$.
- 5: Draw $s_1 \sim \text{Unif}[0.1, 10.0]$ (RNG seed = $42 + r$).
- 6: **for** $t = 1$ to $T_{\text{burn}} + n$, $T_{\text{burn}} = 300$ **do**
- 7: Map s_t to nearest grid point: $i = \text{round}_G(s_t)$.
- 8: Set $a_t = a^*(s_i)$.
- 9: **if** $t > T_{\text{burn}}$ **then**
 store (s_t, a_t) .
- 10: **end if**
- 11: Draw $\varepsilon_{t+1} \sim \mathcal{N}(0, \sigma^2)$ (RNG seed = $42 + r$, sequential draws).
- 12: Set $s_{t+1} = \text{clip}(\rho s_t + w a_t + \varepsilon_{t+1}, 0.1, 10.0)$.
- 13: **end for**
- 14: **return** $(s_t, a_t)_{t=1}^n$.

The DGP is **identical** for Design 1 and Design 2; only the estimators differ.

D.4 ANE-grid estimation

In Design 1, ANE-grid solves the annealed Bellman equation on a coarser grid G with $M = 150$ points and maximises the annealed log-likelihood along a decreasing temperature path.

Algorithm 5 ANE-grid for a single sample $(s_t, a_t)_{t=1}^n$

Require: Data $(s_t, a_t)_{t=1}^n$; sample size n

- 1: Construct grid G , $M = 150$ points on $[0.1, 10.0]$.
- 2: Construct transition matrices P_0, P_1 on G ($N_{\text{sim}} = 2,000$).
- 3: Set $\beta_{\text{rate}} = 0.25$, $\tau_n = n^{-\beta_{\text{rate}}}$. {E.g., $\tau_{5000} = 5000^{-0.25} \approx 0.119$ }
- 4: Construct geometric schedule $\tau_1 = 1.0 > \tau_2 > \dots > \tau_K = \tau_n$ with $K = 8$.
- 5: Initialise $\theta^{(0)} = (1.0, 1.5)$.
- 6: **for** $k = 1$ to K **do**
- 7: Set $\tau = \tau_k$.
- 8: Solve $V_{\theta^{(k-1)}, \tau}$ by value iteration (tolerance 10^{-6} , max 2,000 iterations).
- 9: Compute CCPs $p_{\theta^{(k-1)}, \tau}$ and evaluate $\ell_n(\theta, \tau)$ via interpolation.
- 10: Update $\theta^{(k)} =_{\theta} [-\ell_n(\theta, \tau)]$ using Nelder–Mead, warm-started at $\theta^{(k-1)}$.
- 11: Iteration limit: 400 for $k < K$; 600 for $k = K$. Tolerances: $x_{\text{atol}} = 10^{-5}$, $f_{\text{atol}} = 10^{-6}$.
- 12: **end for**
- 13: **return** $\hat{\theta}_n = \theta^{(K)}$.

Remark D.1 (Feasibility of the annealing rate). *With $\varepsilon_n = 0$ (grid-based), condition (R3) ($\sqrt{n} \varepsilon_n / \tau_n \rightarrow 0$) is vacuous. The remaining conditions require $\beta_{\text{rate}} \in (1/(2\kappa), 1/2)$ (from (R1), (R2), (R4)). With $\beta_{\text{rate}} = 0.25$ and $\kappa > 1$, this is satisfied for $\kappa > 2$ (since $1/(2\kappa) < 0.25$). For the DGP's $\kappa \approx 1$, the condition $1/(2\kappa) = 0.5 > 0.25$ is not satisfied, consistent with the observation in Remark 8.1 that the formal CLT does not directly apply. Nonetheless, $\beta_{\text{rate}} = 0.25$ provides a reasonable practical trade-off between bias (smaller τ_n reduces smoothing bias) and variance (too-small*

τ_n amplifies the score’s sensitivity to near-threshold observations).

D.5 Monte Carlo summary

We repeat the DGP and ANE-grid estimation $R = 200$ times for each $n \in \{1000, 2000, 5000\}$ and compute bias and RMSE for each parameter:

$$\text{Bias}(\hat{\alpha}) = \frac{1}{R} \sum_{r=1}^R (\hat{\alpha}_r - \alpha_0), \quad \text{RMSE}(\hat{\alpha}) = \left(\frac{1}{R} \sum_{r=1}^R (\hat{\alpha}_r - \alpha_0)^2 \right)^{1/2},$$

with analogous definitions for $\hat{\gamma}$. The Monte Carlo standard error of the reported bias is approximately $\text{RMSE}/\sqrt{R}$. Table 3 in Section 8 reports these statistics.

D.6 Hyperparameter summary (Design 1)

Table 5: Hyperparameters for Design 1 (ANE-grid)

Symbol	Value	Description
θ_0	(1.8, 2.0)	True (α, γ)
β	0.95	Discount factor
ρ	0.85	State persistence
w	1.0	Wage
σ	0.4	Transition noise s.d.
$[s_{\min}, s_{\max}]$	[0.1, 10.0]	State space bounds
M_{DGP}	200	DGP grid size
M	150	Estimation grid size
T_{burn}	300	Burn-in periods
β_{rate}	0.25	Annealing rate ($\tau_n = n^{-\beta_{\text{rate}}}$)
K	8	Annealing stages
τ_1	1.0	Initial temperature
N_{sim}	2,000	MC draws for transition matrix
R	200	Monte Carlo replications

E Monte Carlo Design and Algorithms for Design 2

This appendix documents the neural-network experiment (Design 2), in which the value function is approximated by a feedforward network. The DGP is the same hard-max model as in Design 1 (Algorithm 4); only the estimators differ.

As established in Remark 5.5, the neural-network sieve cannot satisfy the feasibility condition (30) for any finite smoothness r when $d_s \geq 1$. Design 2 therefore provides *finite-sample evidence* for the ANE methodology with neural-network value approximation, but falls outside the formal asymptotic theory. Standard errors and coverage are not reported for this reason.

E.1 Neural-network value approximation

Let $V_\phi(s)$ denote a feedforward network with parameter vector $\phi \in \mathbb{R}^p$, mapping $s \in \mathbb{R}$ to $\mathbb{R}$. The architecture is:

$$V_\phi(s) = W^{(4)} \psi(W^{(3)} \psi(W^{(2)} \psi(W^{(1)} s + b^{(1)}) + b^{(2)}) + b^{(3)}) + b^{(4)},$$

where $\psi = \tanh$ is applied componentwise and each hidden layer has width 128. Weights are initialised by Xavier uniform; biases are initialised to zero.

The one-period utility $u(s, a; \theta)$ and transition $s' \sim \rho s + w a + \varepsilon$ are coded as differentiable PyTorch functions. The transition shocks are drawn from $\mathcal{N}(0, \sigma^2)$ on the fly (not pre-generated).

E.2 Neural annealed Bellman operator

For given (θ, τ, ϕ) and state s , the neural approximations to the choice-specific values are

$$\hat{Q}_{\theta, \phi}(s, a) = u(s, a; \theta) + \beta \frac{1}{N_{\text{MC}}} \sum_{k=1}^{N_{\text{MC}}} V_{\phi}(\text{clip}(\rho s + w a + \varepsilon^{(k)}, s_{\min}, s_{\max})),$$

where $\varepsilon^{(k)} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$ and $N_{\text{MC}} = 20$. The annealed value is

$$T_{\theta, \tau} V_{\phi}(s) = \tau \log \left(\exp(\hat{Q}_{\theta, \phi}(s, 0)/\tau) + \exp(\hat{Q}_{\theta, \phi}(s, 1)/\tau) \right).$$

We do **not** solve the fixed-point equation $V = T_{\theta, \tau} V$ exactly. Instead, we choose ϕ to minimise the mean squared Bellman residual over the empirical distribution of states:

$$\mathcal{R}(\phi; \theta, \tau) = \frac{1}{n} \sum_{t=1}^n (V_{\phi}(s_t) - T_{\theta, \tau} V_{\phi}(s_t))^2. \quad (\text{E.1})$$

This is a fitted value iteration (FVI) scheme. For each annealing stage, we update ϕ by $N_V = 80$ steps of Adam (learning rate $\eta_V = 10^{-3}$, batch size $B = 256$). The target $T_{\theta, \tau} V_{\phi}(s)$ is computed with a `no_grad` context to stabilise training (semi-gradient update).

E.3 Neural annealed choice probabilities and likelihood

Given (θ, τ, ϕ) , the annealed choice probability at state s is

$$p_{\theta, \phi, \tau}(1 | s) = \frac{\exp(\hat{Q}_{\theta, \phi}(s, 1)/\tau)}{\exp(\hat{Q}_{\theta, \phi}(s, 0)/\tau) + \exp(\hat{Q}_{\theta, \phi}(s, 1)/\tau)},$$

clipped to $[10^{-7}, 1 - 10^{-7}]$. The annealed log-likelihood is

$$\ell_n(\theta, \phi, \tau) = \sum_{t=1}^n [a_t \log p_{\theta, \phi, \tau}(1 | s_t) + (1 - a_t) \log(1 - p_{\theta, \phi, \tau}(1 | s_t))].$$

E.4 ANE-NN algorithm

Algorithm 6 ANE-NN for a single sample $(s_t, a_t)_{t=1}^n$

Require: Data $(s_t, a_t)_{t=1}^n$; sample size n

- 1: Initialise $\theta^{(0)} = (1.0, 1.5)$.
 - 2: Initialise value network $V_{\phi^{(0)}}$ (Xavier uniform).
 - 3: Construct geometric schedule $\tau_1 = 1.0 > \dots > \tau_K = n^{-0.30}$, $K = 8$.
 - 4: **for** $k = 1$ to K **do**
 - 5: Set $\tau = \tau_k$.
 - 6: **Value-step.** Freeze $\theta^{(k-1)}$. Update ϕ by $N_V = 80$ Adam steps on $\mathcal{R}(\phi; \theta^{(k-1)}, \tau)$ (learning rate $\eta_V = 10^{-3}$, batch size $B = 256$, $N_{MC} = 20$). Denote result $\phi^{(k)}$.
 - 7: **Parameter-step.** Freeze $V_{\phi^{(k)}}$ (set `requires_grad=False`). Update θ by $N_\theta = 60$ Adam steps on $-\ell_n(\theta, \phi^{(k)}, \tau)$ with mini-batches of size $B = 256$.
 - Learning rate: $\eta_\theta = 3 \times 10^{-3} \cdot (\tau/\tau_1)^{0.3}$.
 - Gradient clipping: $\|\nabla_\theta\|_2 \leq 1.0$.
 - 8: Denote result $\theta^{(k)}$.
 - 9: Unfreeze $V_{\phi^{(k)}}$ (set `requires_grad=True`).
 - 10: **end for**
 - 11: **return** $\hat{\theta}_n^{\text{ANE-NN}} = \theta^{(K)}$.
-

Remark E.1 (Learning rate schedule). *The decaying learning rate $\eta_\theta \propto (\tau/\tau_1)^{0.3}$ reflects the fact that the annealed likelihood becomes sharper as $\tau \rightarrow 0$: the Lipschitz constant of the score scales as $O(1/\tau)$ (Lemma 4.5), so smaller steps are needed to avoid overshooting. The exponent 0.3 was chosen by preliminary experiments.*

E.5 Fixed-soft-max NPL-NN

Nested pseudo-likelihood with neural value approximation (NPL-NN) uses a **fixed** soft-max temperature $\tau \in \{0.5, 0.3, 0.1\}$ throughout estimation. Because τ does not vanish with n , NPL-NN converges to a pseudo-true parameter $\tilde{\theta}(\tau) \neq \theta_0$ whenever the DGP is hard-max.

Algorithm 7 NPL-NN with fixed τ

Require: Data $(s_t, a_t)_{t=1}^n$; fixed $\tau \in \{0.5, 0.3, 0.1\}$

- 1: Initialise $\theta^{(0)} = (1.0, 1.5)$; initialise $V_{\phi^{(0)}}$.
 - 2: **for** $m = 0$ to $M_{\text{outer}} - 1 = 4$ **do**
 - 3: **Value-step.** Freeze $\theta^{(m)}$. Update ϕ by $N_V = 80$ Adam steps on $\mathcal{R}(\phi; \theta^{(m)}, \tau)$.
 - 4: **Parameter-step.** Freeze $V_{\phi^{(m)}}$. Update θ by $N_\theta = 60$ Adam steps on $-\ell_n(\theta, \phi^{(m)}, \tau)$ (learning rate $\eta_\theta = 3 \times 10^{-3}$).
 - 5: **end for**
 - 6: **return** $\hat{\theta}_n^{\text{NPL-NN}}(\tau) = \theta^{(M_{\text{outer}})}$.
-

The architecture, batch size, and number of MC draws ($N_{MC} = 20$) are identical to those of ANE-NN. The only difference is that the temperature is fixed rather than annealed.

E.6 CCP-NN

The CCP-NN estimator is a neural analogue of two-step CCP methods (Hotz and Miller, 1993).

Algorithm 8 CCP-NN: two-stage estimator

Require: Data $(s_t, a_t)_{t=1}^n$

- 1: **First stage (CCP network).** Fit a feedforward network $p_\psi(s)$ (same architecture as V_ϕ but with sigmoid output) to approximate $\Pr(a = 1 \mid s)$ by minimising the binary cross-entropy:

$$\mathcal{L}(\psi) = -\frac{1}{n} \sum_{t=1}^n [a_t \log p_\psi(s_t) + (1 - a_t) \log(1 - p_\psi(s_t))].$$

Train for $N_{\text{CCP}} = 400$ Adam steps (learning rate 10^{-3} , batch size 256). Freeze p_ψ after training.

- 2: **Second stage (structural logit).** Fix a structural temperature $\tau_{\text{struct}} = 0.5$. Define the flow utility difference

$$\Delta u_\theta(s) = u(s, 1; \theta) - u(s, 0; \theta).$$

Estimate θ by maximising the *static structural log-likelihood*:

$$\max_{\theta} \frac{1}{n} \sum_{t=1}^n [a_t \log \sigma(\Delta u_\theta(s_t)/\tau_{\text{struct}}) + (1 - a_t) \log(1 - \sigma(\Delta u_\theta(s_t)/\tau_{\text{struct}}))], \quad (\text{E.2})$$

where $\sigma(\cdot)$ is the logistic sigmoid. Optimise by $N_{\theta, \text{CCP}} = 250$ Adam steps (learning rate 5×10^{-3} , batch size 256), starting from $\theta = (1.0, 1.5)$.

- 3: **return** $\hat{\theta}_n^{\text{CCP-NN}} = \theta$.
-

Remark E.2 (Static approximation in CCP-NN). *The second stage (E.2) ignores the continuation value entirely and matches only the static flow-utility difference. This is a deliberate simplification: the first-stage CCP network absorbs the continuation-value differences into the estimated probabilities, and the structural logit “inverts” these probabilities back to structural parameters. In practice, this approach is ill-suited to hard-max environments because the first-stage network smooths the sharp policy discontinuity, causing the second-stage inversion to produce severely biased estimates (see Table 4).*

E.7 Monte Carlo implementation

For each design point $(n, \text{estimator})$ we run $R = 200$ replications. For each replication r we:

- (i) Simulate $(s_t, a_t)_{t=1}^n$ from the hard-max DGP at θ_0 via Algorithm 4 (seed = 42 + r).
- (ii) Compute $\hat{\theta}_r^{\text{ANE-NN}}$ using Algorithm 6 with $\tau_n = n^{-0.30}$.
- (iii) Compute $\hat{\theta}_r^{\text{NPL-NN}}(\tau)$ for each $\tau \in \{0.5, 0.3, 0.1\}$ using Algorithm 7.
- (iv) Compute $\hat{\theta}_r^{\text{CCP-NN}}$ via Algorithm 8.

For each estimator and parameter $p \in \{\alpha, \gamma\}$, we report Monte Carlo bias and RMSE:

$$\text{Bias}(\hat{p}) = \frac{1}{R} \sum_{r=1}^R (\hat{p}_r - p_0), \quad \text{RMSE}(\hat{p}) = \left(\frac{1}{R} \sum_{r=1}^R (\hat{p}_r - p_0)^2 \right)^{1/2}.$$

Standard errors and confidence-interval coverage are not reported for Design 2, because the neural-network sieve does not satisfy the rate conditions (R1)–(R4) simultaneously (Remark 5.5), and the sandwich variance estimator from Section 6.5 is not theoretically justified.

E.8 Hyperparameter summary (Design 2)

Table 6: Hyperparameters for Design 2 (neural-network estimators)

Symbol	Value	Description
<i>Common</i>		
θ_0	(1.8, 2.0)	True (α, γ)
Network	3×128 Tanh	Hidden layers $\times$ width
N_{MC}	20	MC draws for $\mathbb{E}[V(s')]$
B	256	Mini-batch size
R	200	Monte Carlo replications
<i>ANE-NN</i>		
β_{rate}	0.30	Annealing rate ($\tau_n = n^{-0.30}$)
K	8	Annealing stages
τ_1	1.0	Initial temperature
N_V	80	Value-step Adam iterations
η_V	10^{-3}	Value-step learning rate
N_θ	60	θ -step Adam iterations
η_θ	$3 \times 10^{-3} \cdot (\tau/\tau_1)^{0.3}$	θ -step learning rate
Gradient clip	$\ \nabla\ _2 \leq 1.0$	Applied to θ updates
<i>NPL-NN</i>		
M_{outer}	5	Policy iterations
τ	{0.5, 0.3, 0.1}	Fixed temperatures
<i>CCP-NN</i>		
N_{CCP}	400	First-stage training steps
$N_{\theta, CCP}$	250	Second-stage training steps
τ_{struct}	0.5	Structural logit temperature
η_{CCP}	$10^{-3} / 5 \times 10^{-3}$	Learning rates (stage 1 / stage 2)

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